Pipeline - ONLY dataset - Benchmark summary

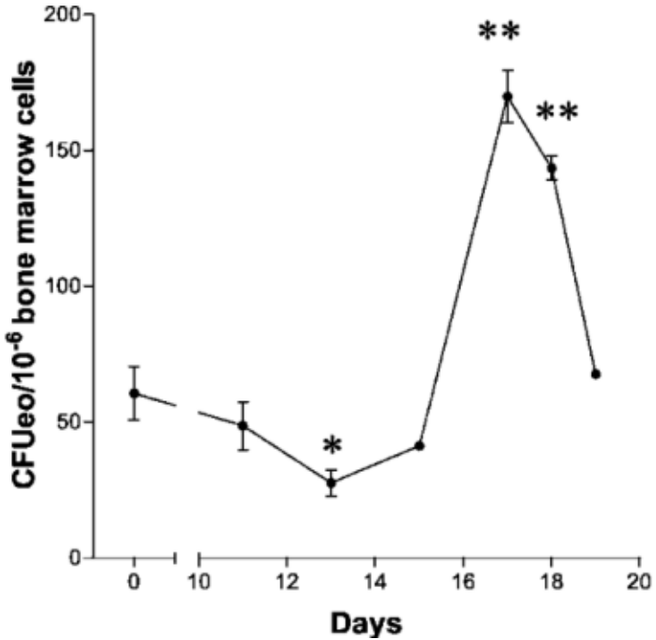
Model: gemini-3-pro-preview
Cases: 35
Successes: 34

nMAE (mean / median, n=34) 0.0761 / 0.0367
L1 normalized (mean / median, n=33) 0.0412 / 0.0279
Duration seconds (mean / median, n=34) 16.6 / 16.9
Series count mismatches (extracted vs GT): 0

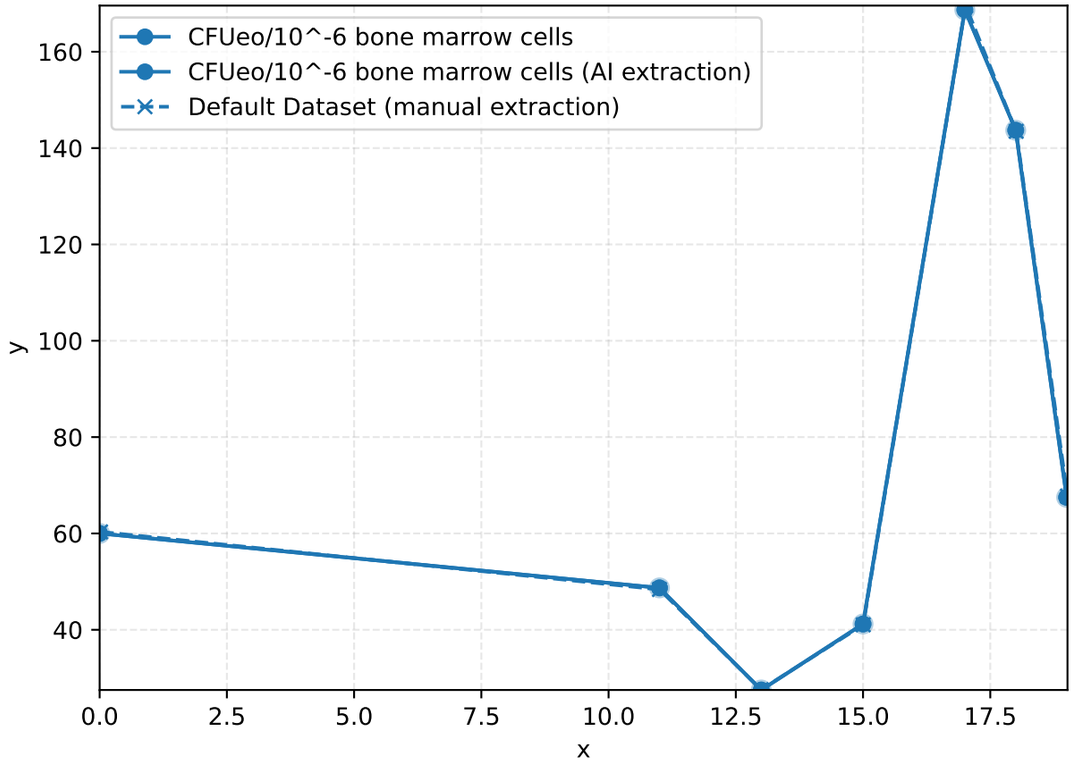
Pipeline - ONLY dataset - Benchmark summary - Prompt

```
 You are an expert system that extracts NUMERICAL data from 2D scientific plots. You must
return ALL visible series from the given image.  Rules: - Do not guess or invent data. -
Work in three internal steps (do NOT show them):   1) Read axes:      - Determine x_min,
x_max (detect log10 if any).      - Determine y_min, y_max (detect log10 if any).      -
List visible tick labels for x and y (as numbers).   2) Identify all visible data series:
- For discrete markers: read their positions and map to axis coordinates.      - For
continuous curves: sample >= 20 points (include local extrema, inflections, and uniform
samples).   3) Validate:      - Ensure x in [x_min, x_max], y in [y_min, y_max].      - For
linear axes, reject perfectly uniform sequences unless they exactly match tick labels.
- If uncertain about a point, omit it. Never invent.  Output format (STRICT JSON, no extra
text): {   "series": [     {       "label": "<legend name if visible else 'series_N'>",
"points": [         {"x": <float>, "y": <float>, "confidence": <float 0..1>},         ...
],       "axis": {         "x": {"min": <float>, "max": <float>, "scale": "linear"|"log10"},
"y": {"min": <float>, "max": <float>, "scale": "linear"|"log10"}       }     },     ...   ]
}  Additional rules: - Include ALL visible series. - Keep points sorted by increasing x. -
Use at least 20 points for continuous curves if readable; otherwise return all discrete
markers. - Never include explanations, placeholders, or text outside JSON.
```

Source image



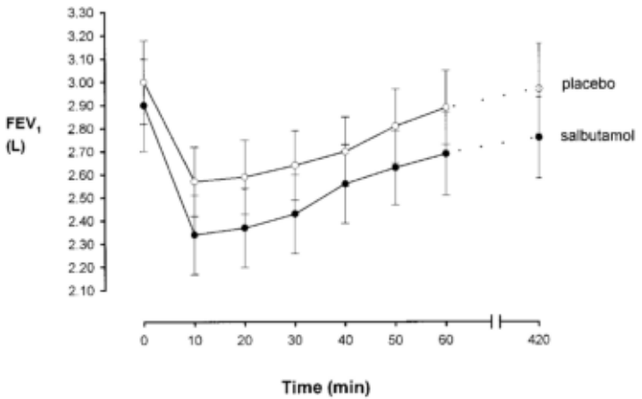
Data-as-cfVO-vMjk



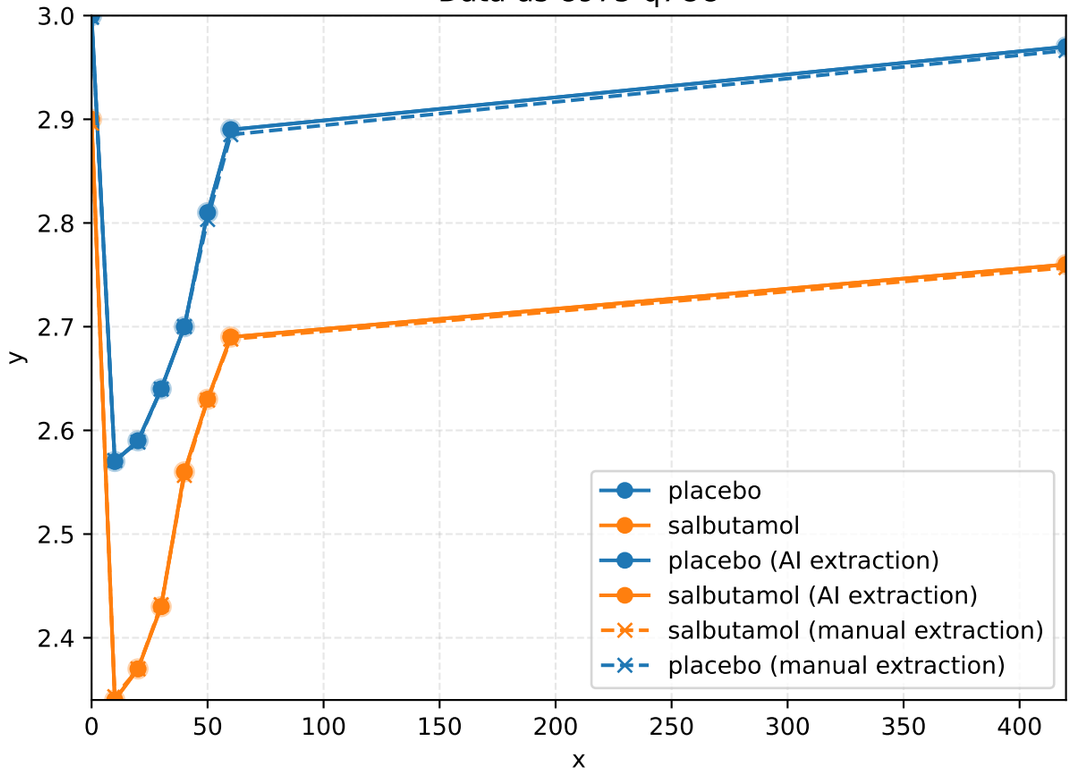
Metrics

Metric	Value
duration (s)	6
overall_nmae	0.002682
overall_integral_l1	1.863
overall_integral_l1_normalized	0.0006913
overall_x_misaligned	False
extracted_series_count	1
GT_series_count	1

Source image



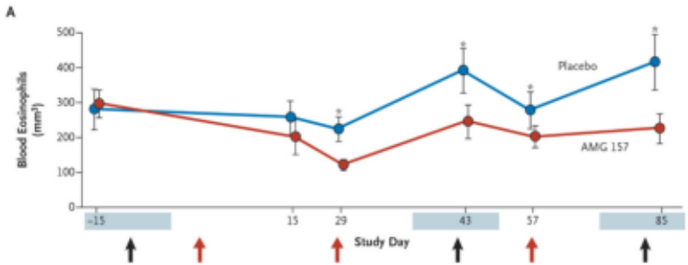
Data-as-8975-q7O8



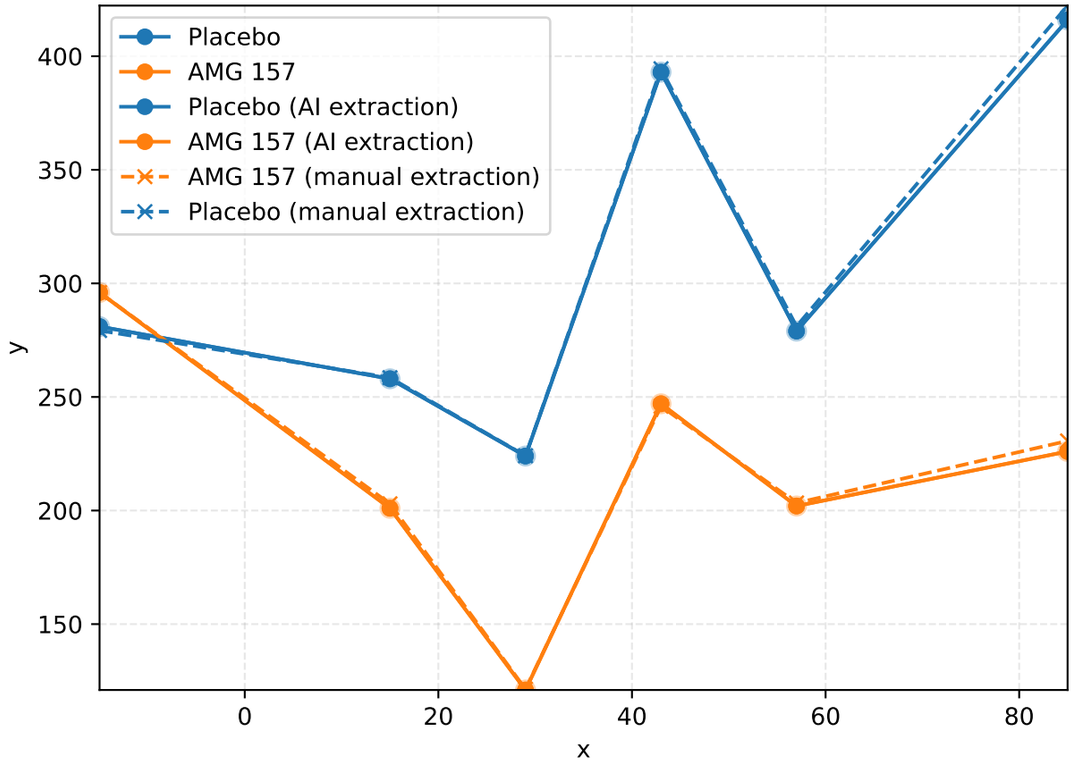
Metrics

Metric	Value
duration (s)	9
overall_nmae	0.004978
overall_integral_l1	1.363
overall_integral_l1_normalized	0.006927
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



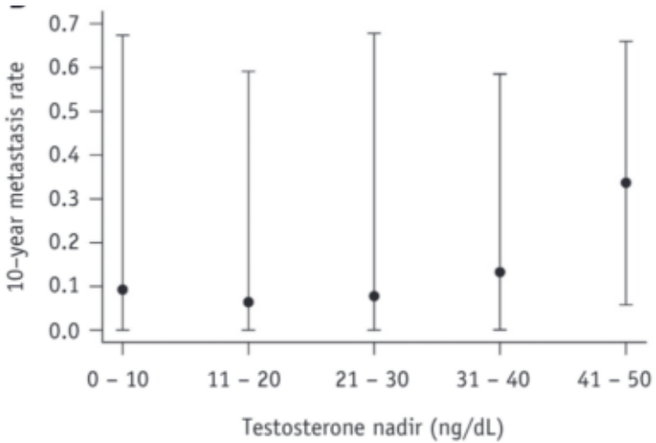
Data-as-8IYI-AGTa



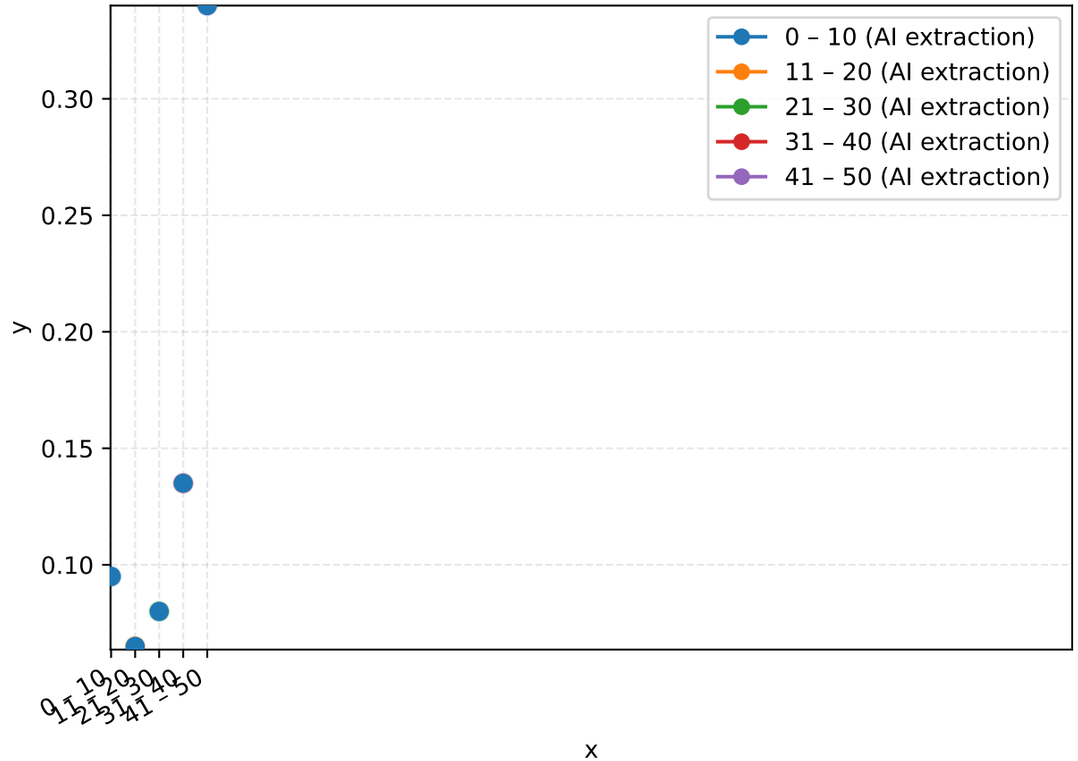
Metrics

Metric	Value
duration (s)	7
overall_nmae	0.009643
overall_integral_l1	174.5
overall_integral_l1_normalized	0.009344
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



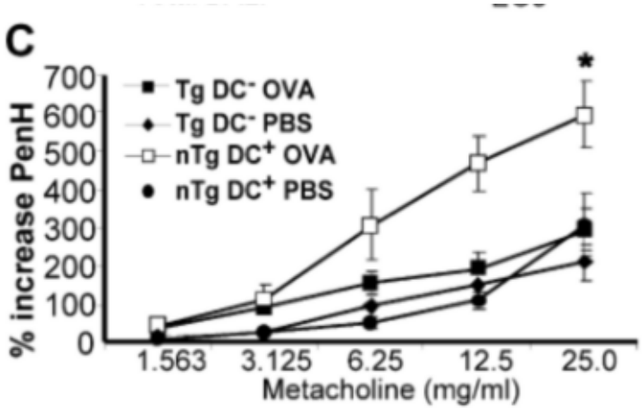
Data-ProstateCancer-10YearMetastasis-TestosteroneNadir-as-HUFM-k4RT

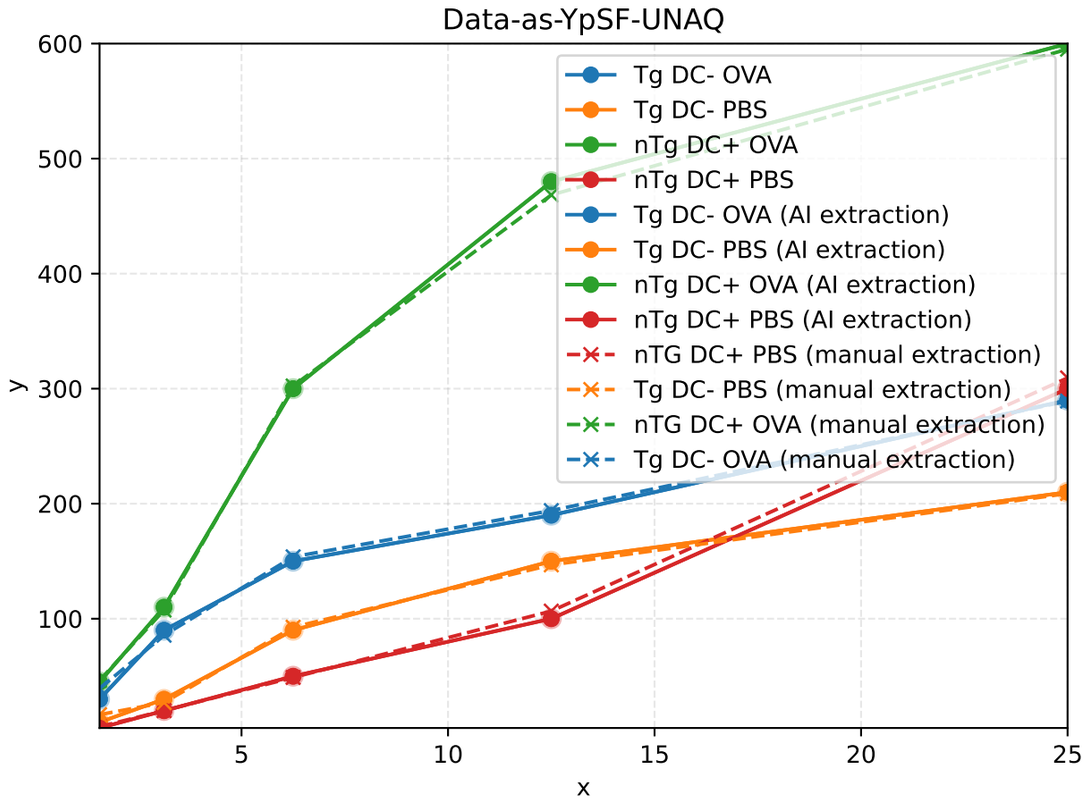


Metrics

Metric	Value
duration (s)	8
overall_nmae	0.01086
overall_integral_l1	nan
overall_integral_l1_normalized	—
overall_x_misaligned	True
extracted_series_count	5
GT_series_count	5

Source image

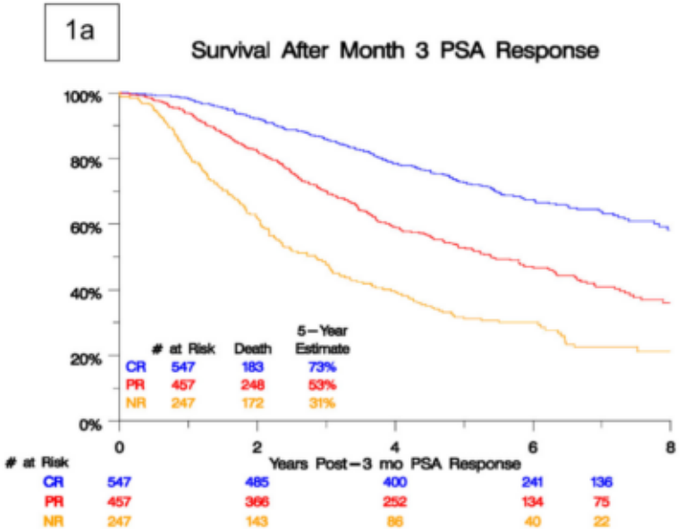


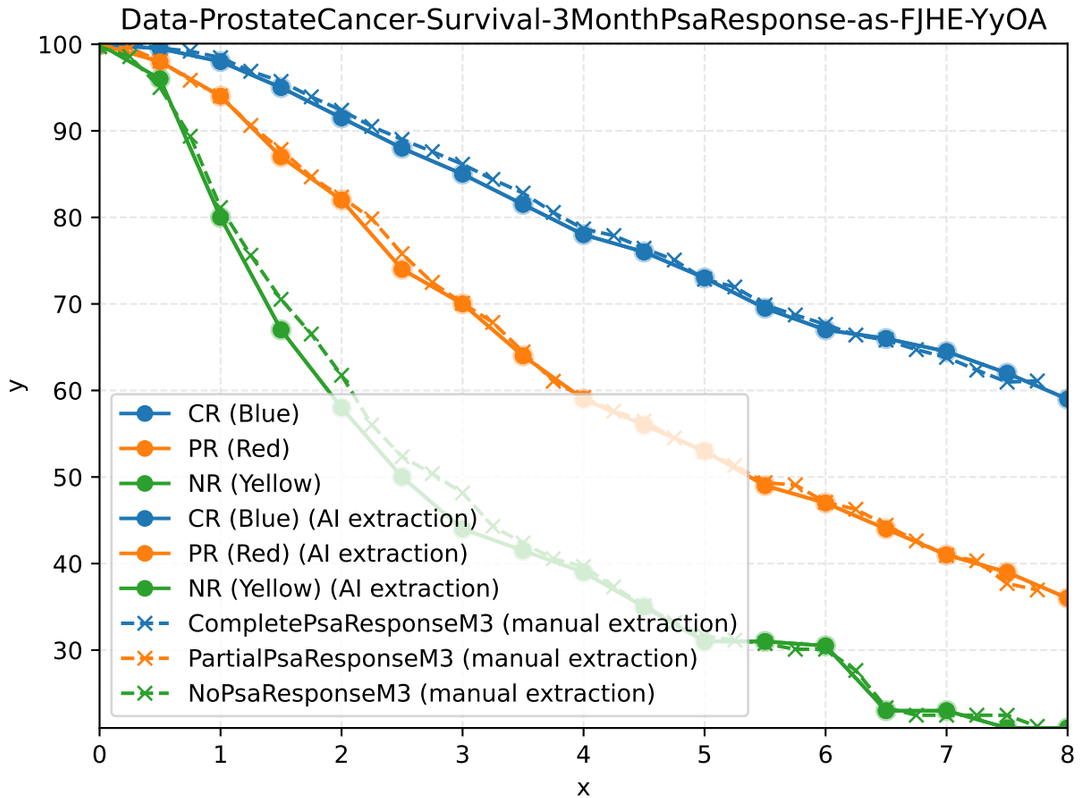


Metrics

Metric	Value
duration (s)	11
overall_nmae	0.01343
overall_integral_l1	105.6
overall_integral_l1_normalized	0.01421
overall_x_misaligned	False
extracted_series_count	4
GT_series_count	4

Source image

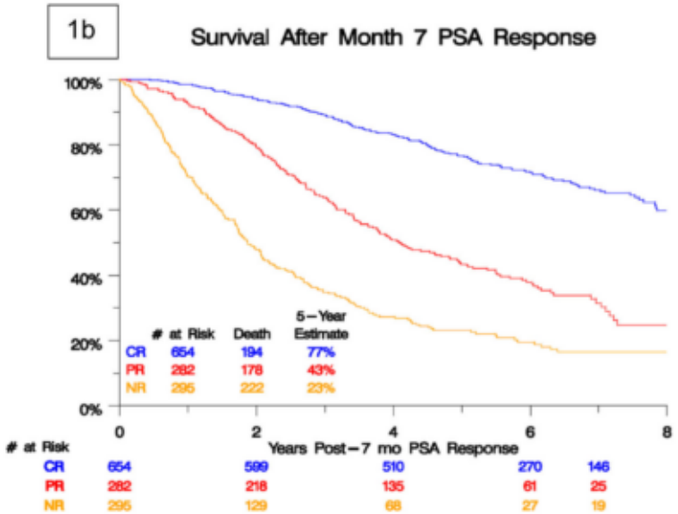


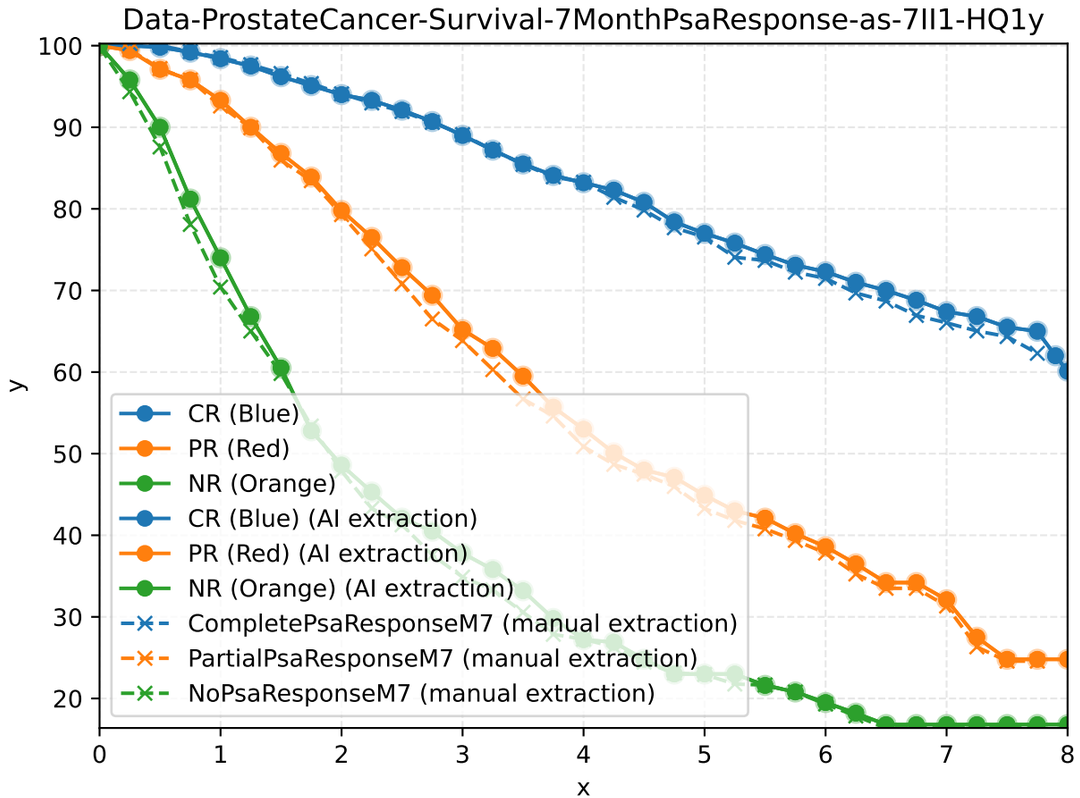


Metrics

Metric	Value
duration (s)	18
overall_nmae	0.01352
overall_integral_l1	5.957
overall_integral_l1_normalized	0.01289
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image

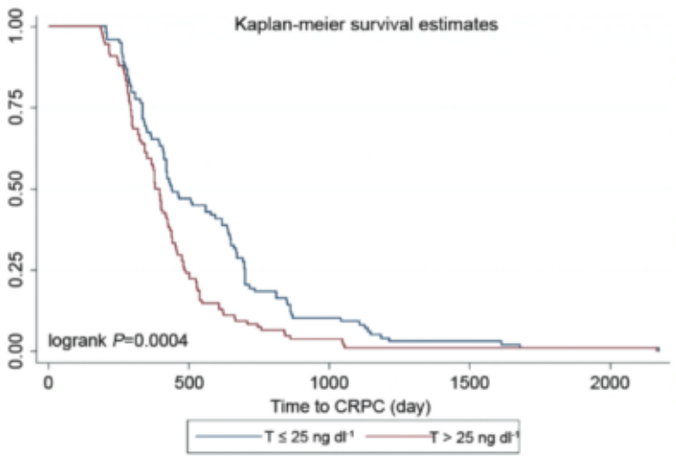




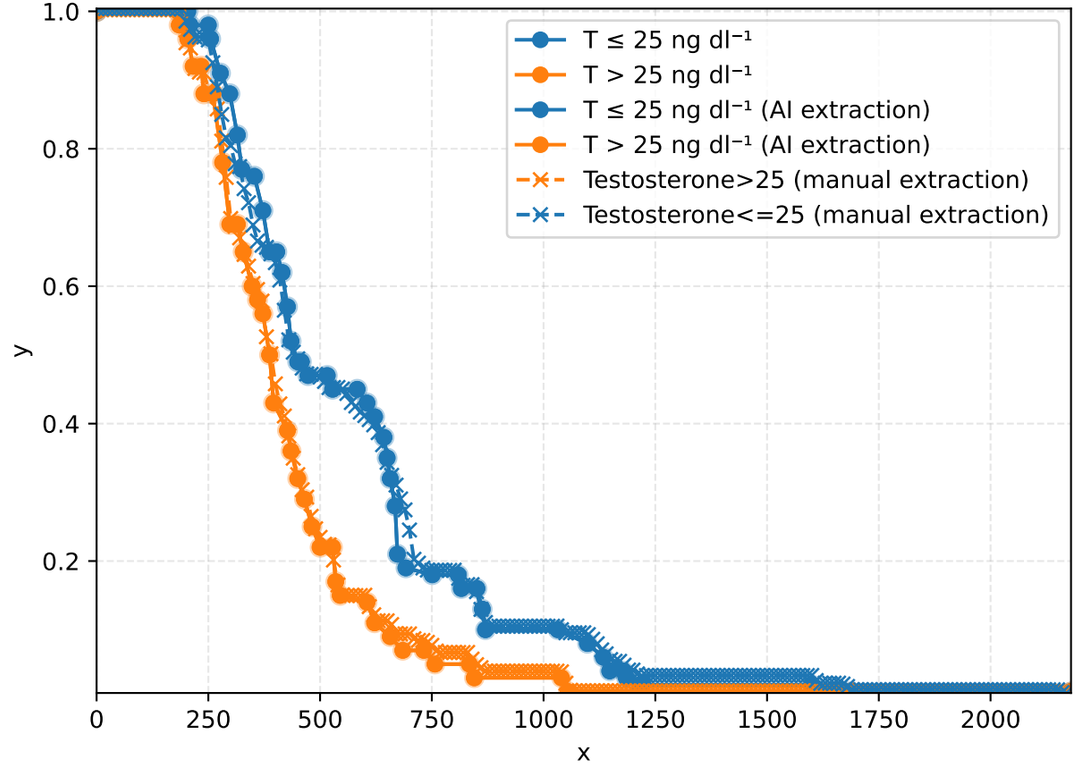
Metrics

Metric	Value
duration (s)	29
overall_nmae	0.01571
overall_integral_l1	7.241
overall_integral_l1_normalized	0.01502
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



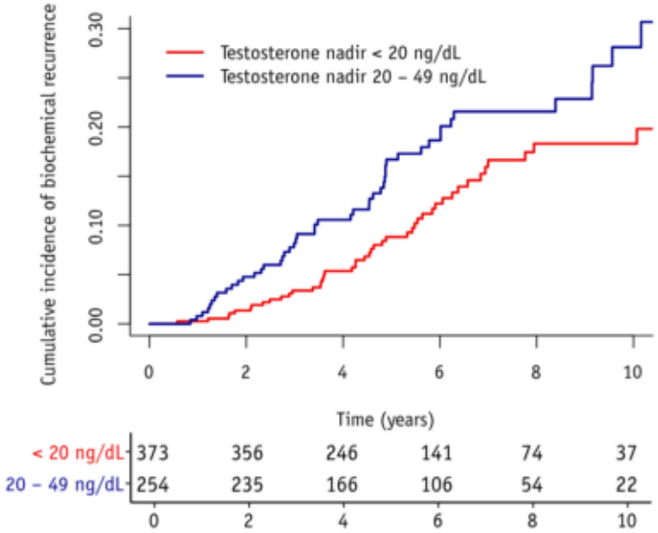
Data-ProstateCancer-TimeToCastrationResistance-CastrationLevel-as-G8iP-2p8H



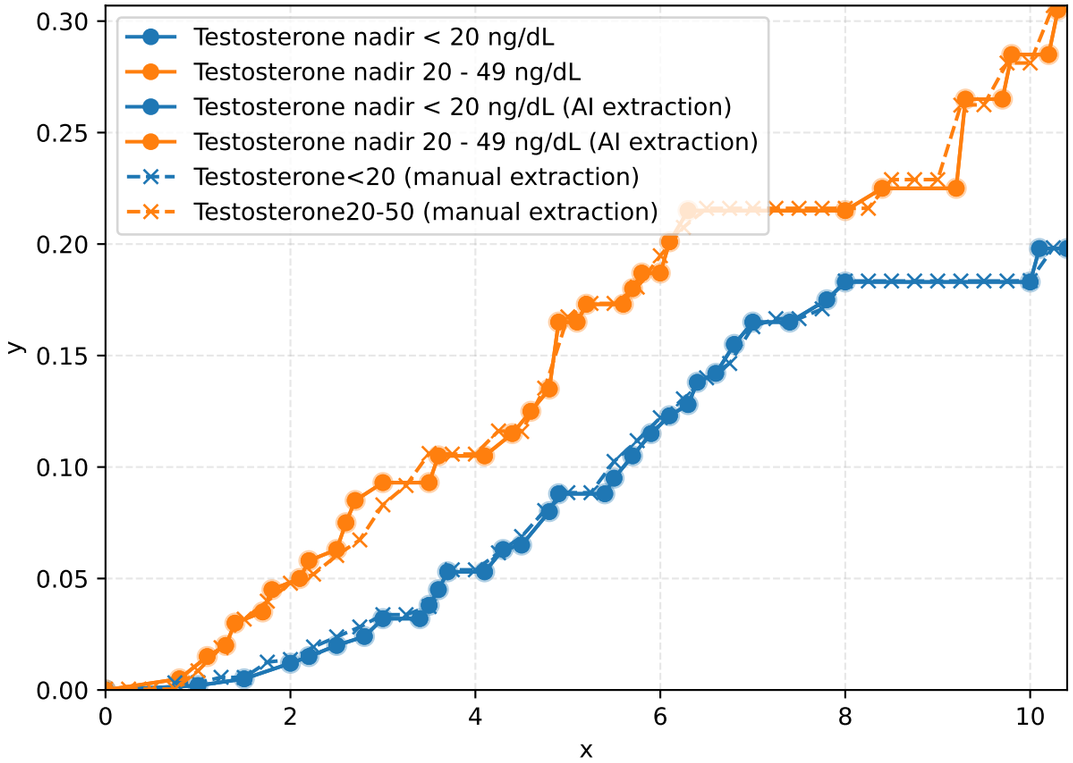
Metrics

Metric	Value
duration (s)	27
overall_nmae	0.01711
overall_integral_l1	18.85
overall_integral_l1_normalized	0.008727
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



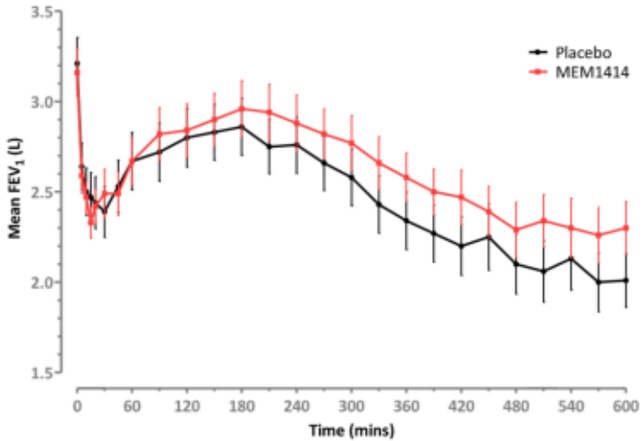
Data-ProstateCancer-BiochemicalRecurrence-TestosteroneNadir-as-kPUM-ja2E



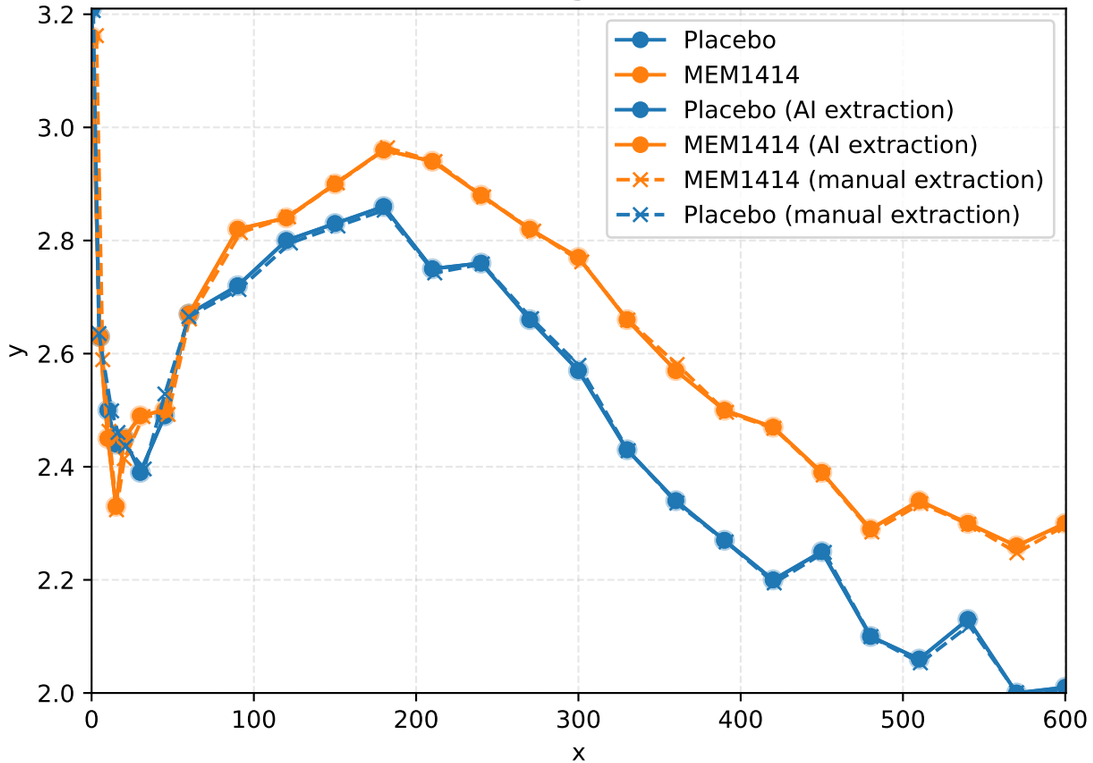
Metrics

Metric	Value
duration (s)	23
overall_nmae	0.01719
overall_integral_l1	0.03961
overall_integral_l1_normalized	0.01497
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



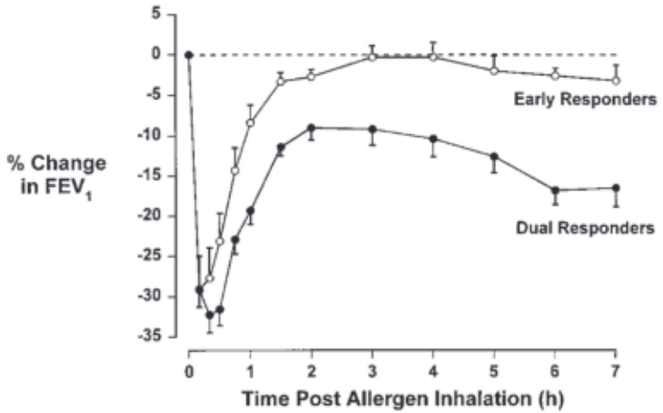
Data-as-ngkV-QcWm



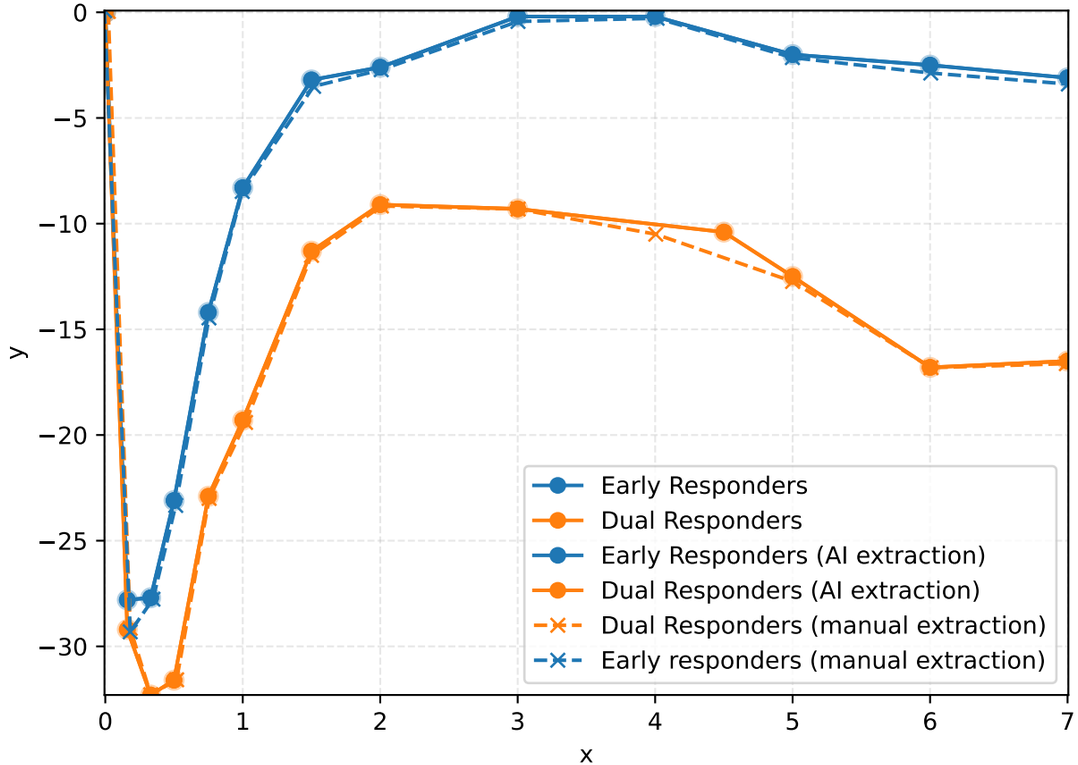
Metrics

Metric	Value
duration (s)	21
overall_nmae	0.01743
overall_integral_l1	3.656
overall_integral_l1_normalized	0.005823
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



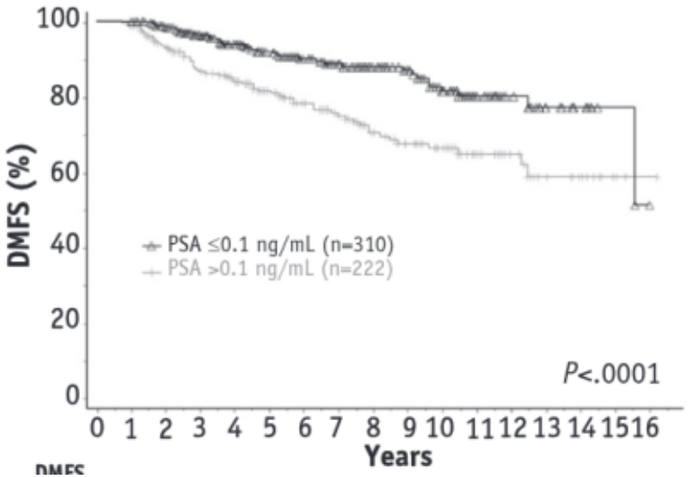
Data-as-bU4o-xO6R



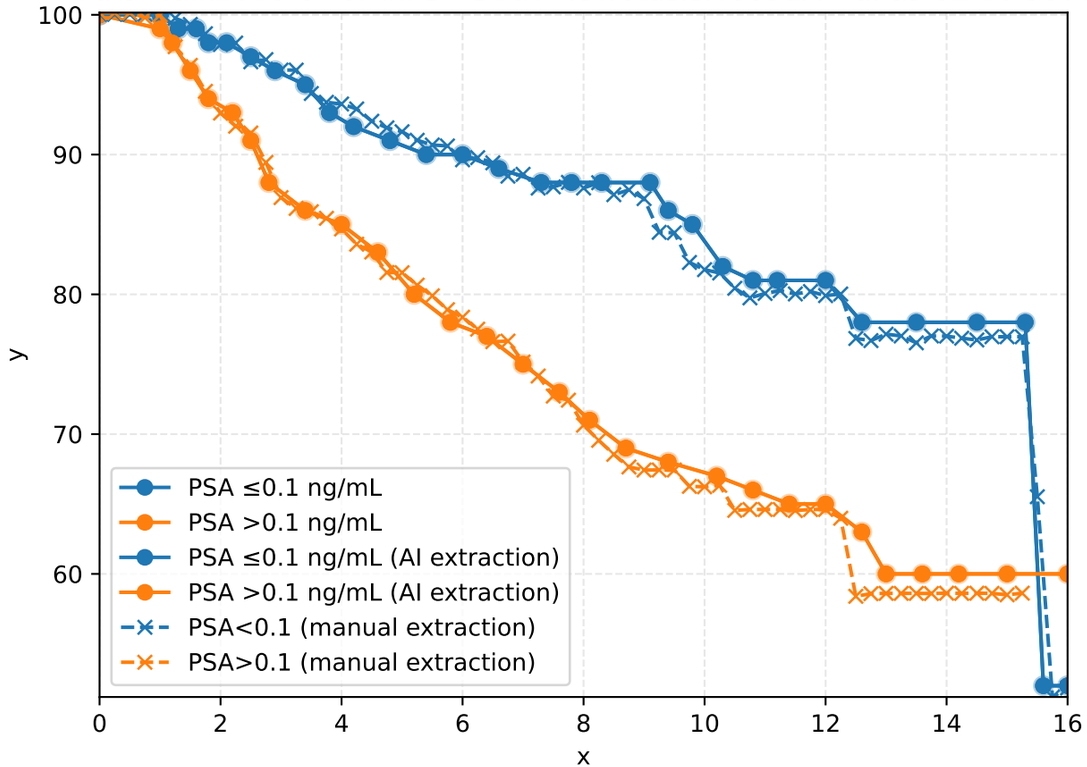
Metrics

Metric	Value
duration (s)	11
overall_nmae	0.02059
overall_integral_l1	2.076
overall_integral_l1_normalized	0.009633
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



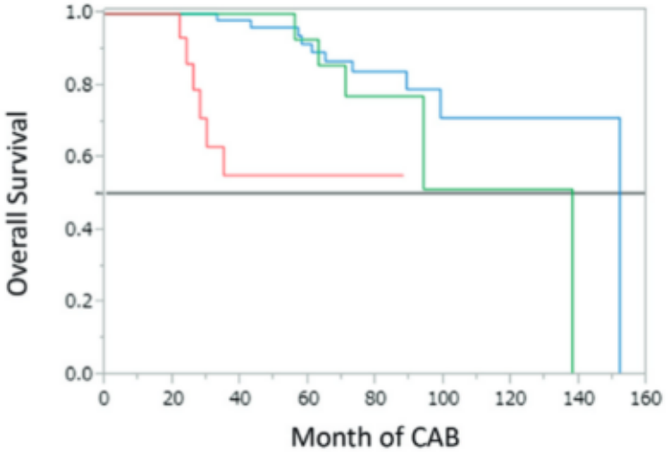
Data-Naik2016-6MonthPSA-Strat-MetastasisFreeSurvival-as-cLMA-G8dv



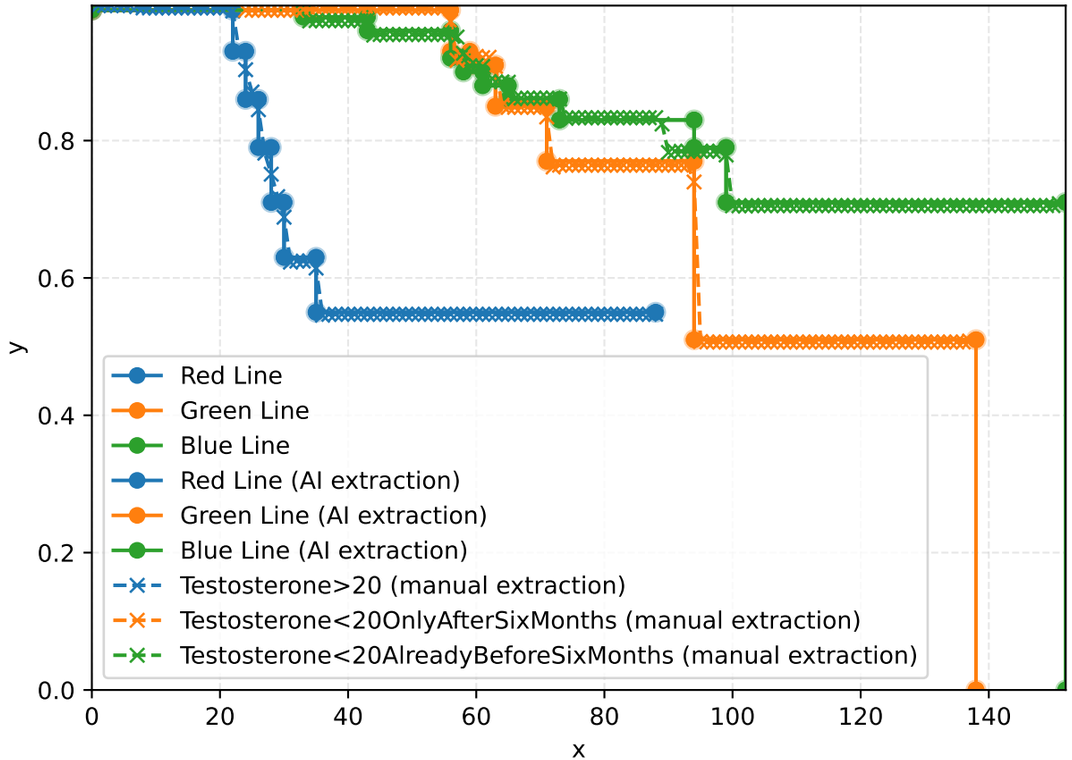
Metrics

Metric	Value
duration (s)	19
overall_nmae	0.02099
overall_integral_l1	14.17
overall_integral_l1_normalized	0.01983
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



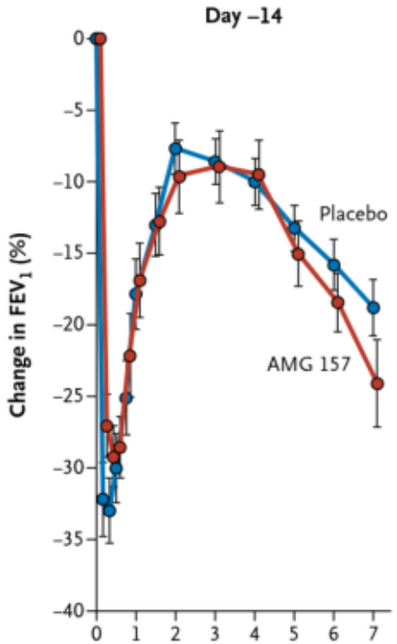
Data-ProstateCancer-OverallSurvival-TestosteroneResponseNadir-as-7x4S-GPr7



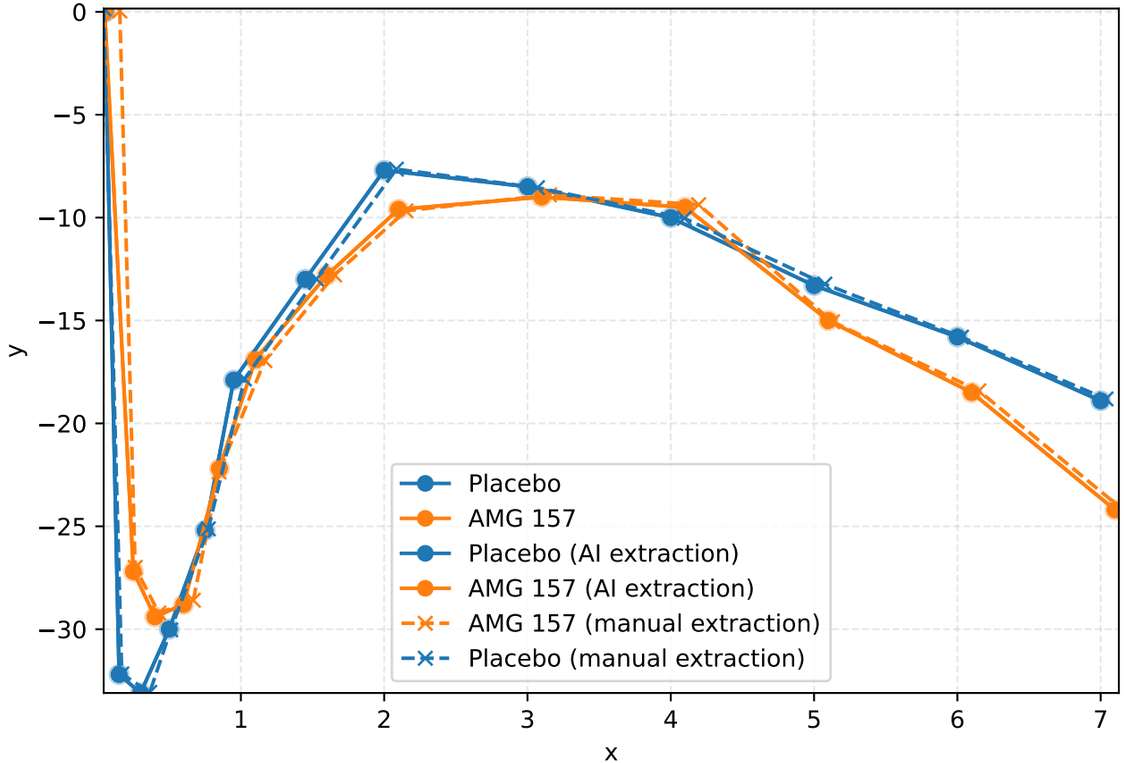
Metrics

Metric	Value
duration (s)	19
overall_nmae	0.02722
overall_integral_l1	0.5608
overall_integral_l1_normalized	0.01443
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



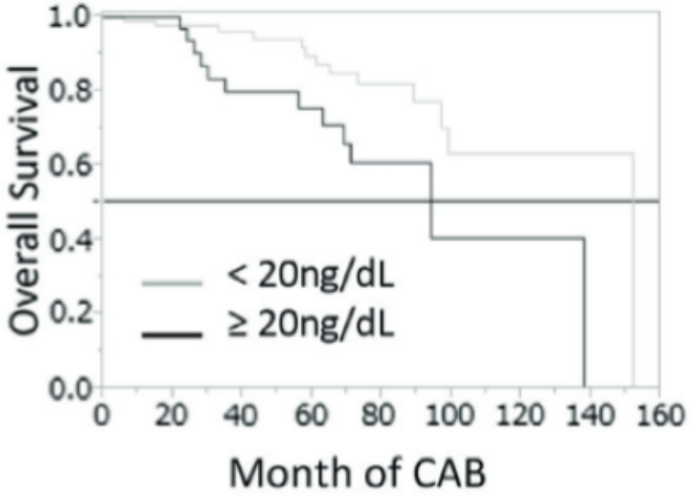
Data-as-b98m-TsDT



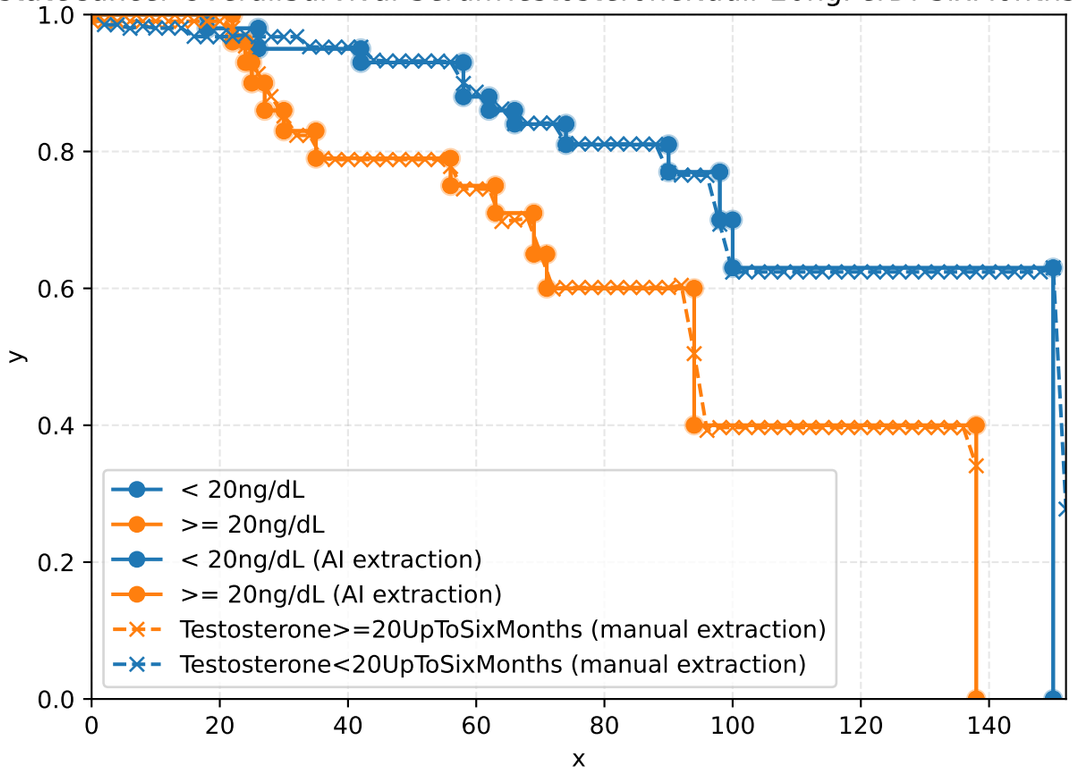
Metrics

Metric	Value
duration (s)	11
overall_nmae	0.02796
overall_integral_l1	2.651
overall_integral_l1_normalized	0.0121
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



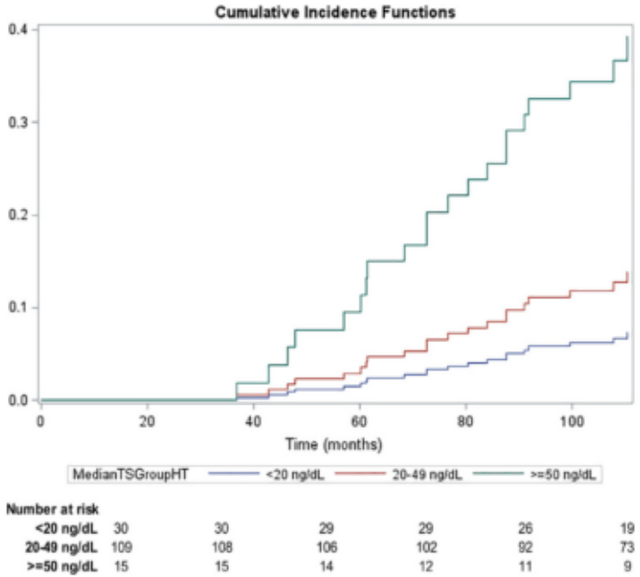
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-20ngPerDl-SixMonths-as-Byus-Y7YM



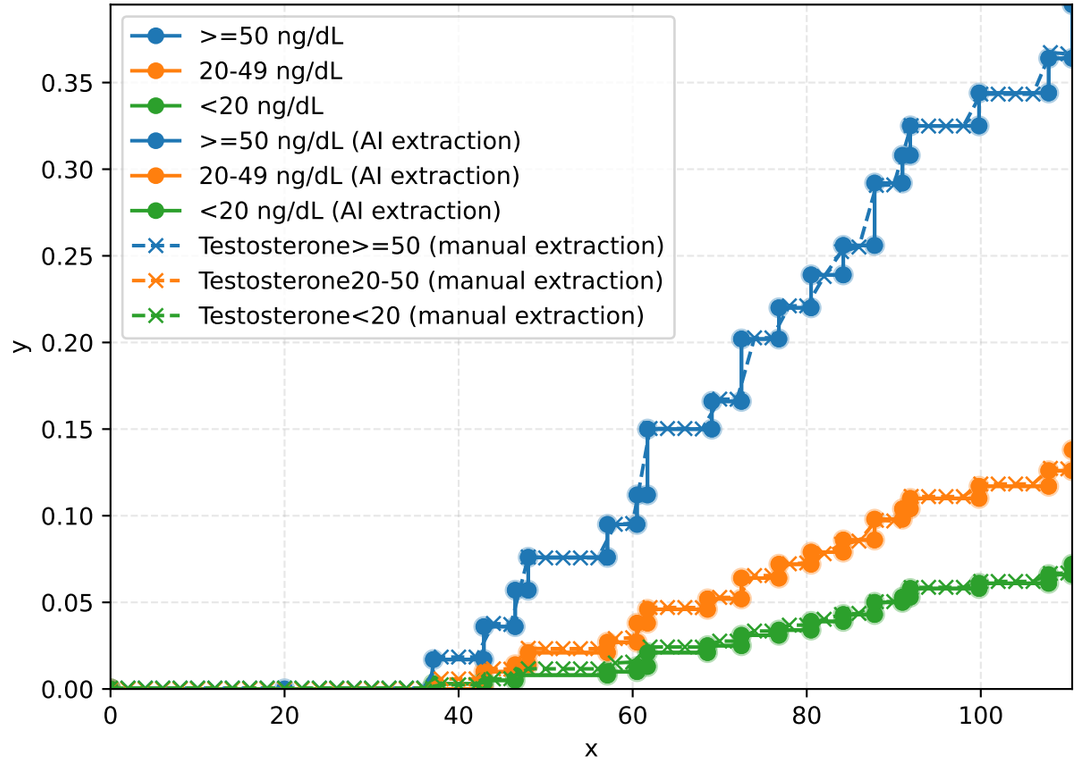
Metrics

Metric	Value
duration (s)	17
overall_nmae	0.03168
overall_integral_l1	2.935
overall_integral_l1_normalized	0.03111
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



Data-ProstateCancer-TimeToBiochemicalFailure-TestosteroneMedian-as-V4ML-TC46

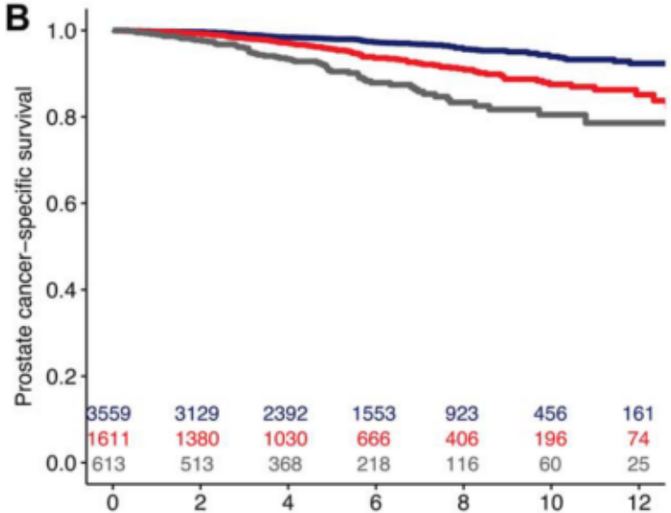


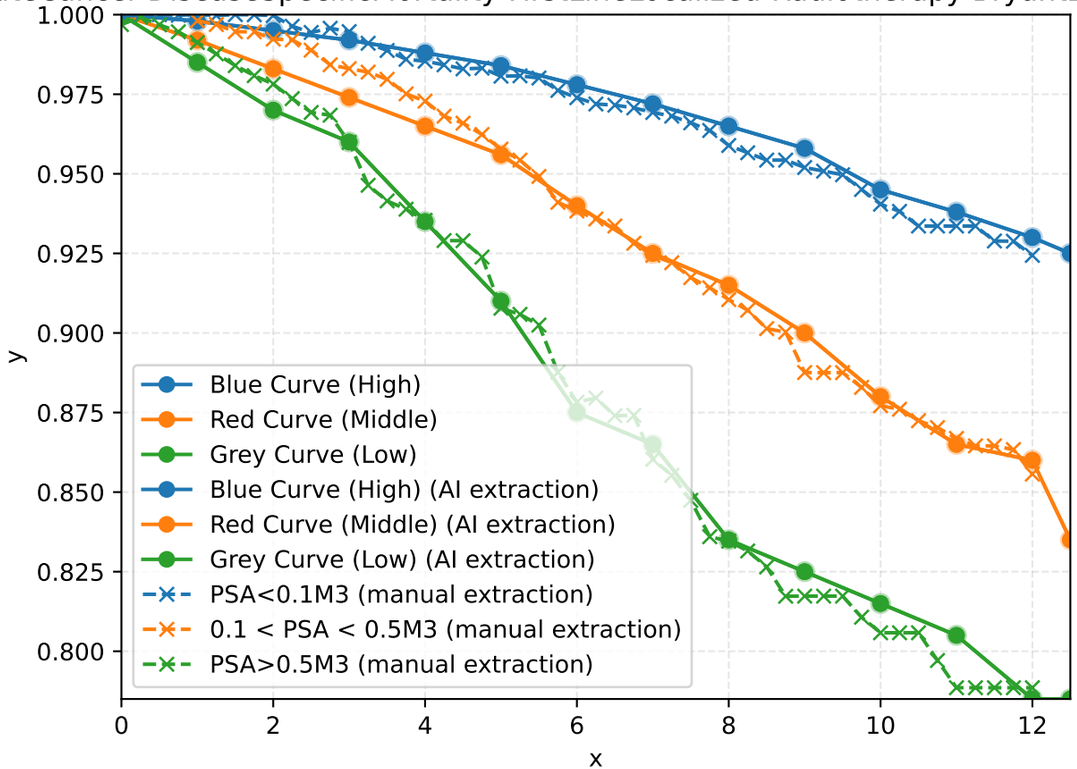
Metrics

Metric	Value
duration (s)	32
overall_nmae	0.03611
overall_integral_l1	0.4531
overall_integral_l1_normalized	0.02787
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Data-ProstateCancer-DiseaseSpecificMortality-FirstLineLocalized-Radiotherapy-Bryant2018-as-PBP9-LpbG

Source image

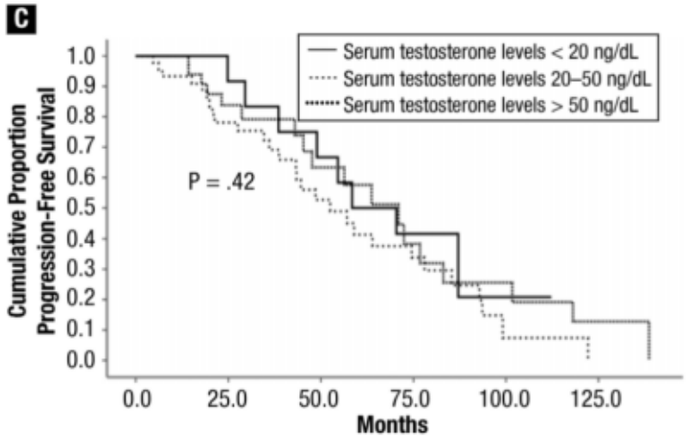




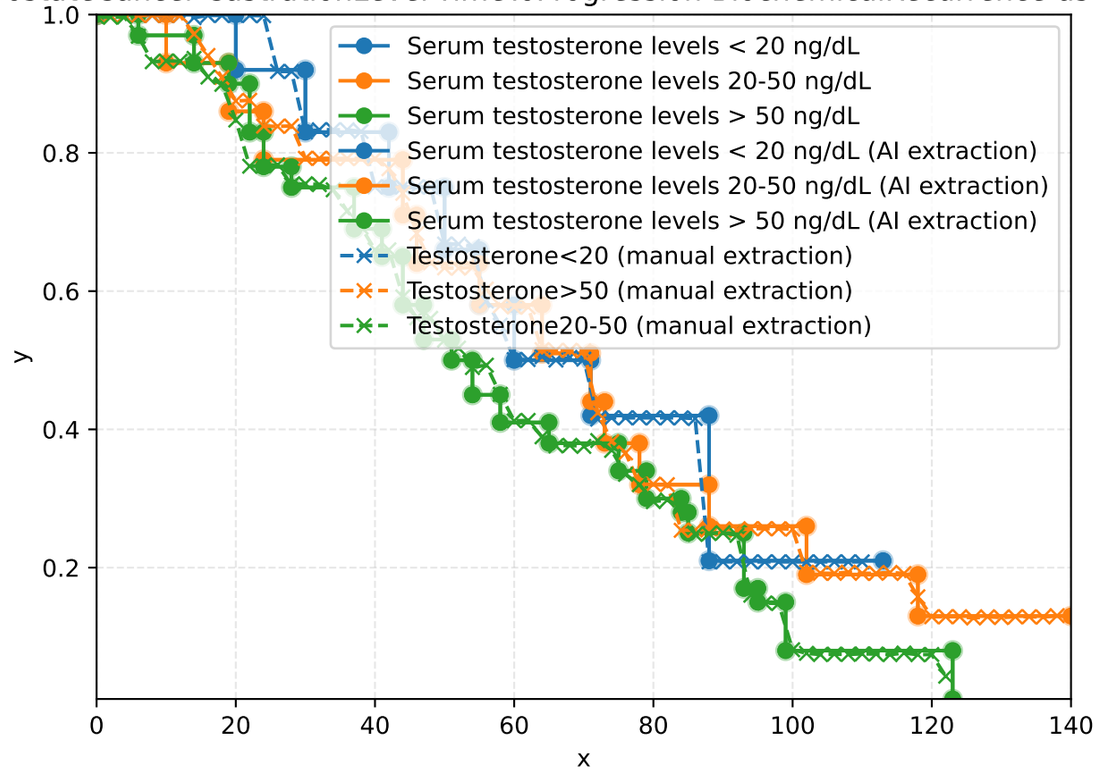
Metrics

Metric	Value
duration (s)	17
overall_nmae	0.03663
overall_integral_l1	0.05518
overall_integral_l1_normalized	0.03732
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



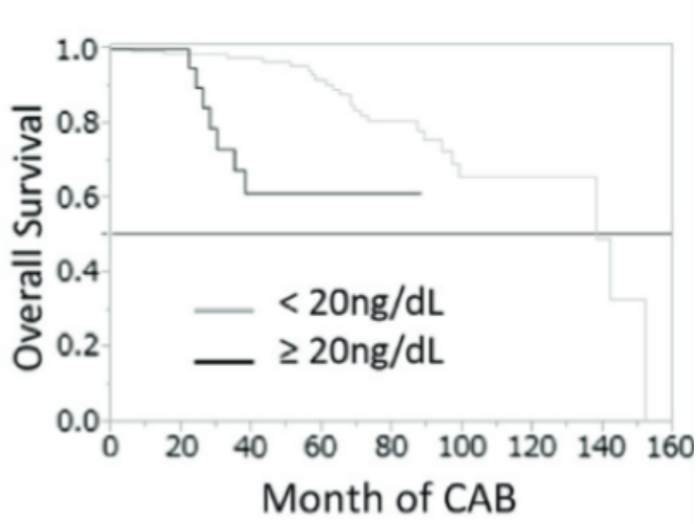
Data-ProstateCancer-CastrationLevel-TimeToProgression-BiochemicalRecurrence-as-4dag-RoWD



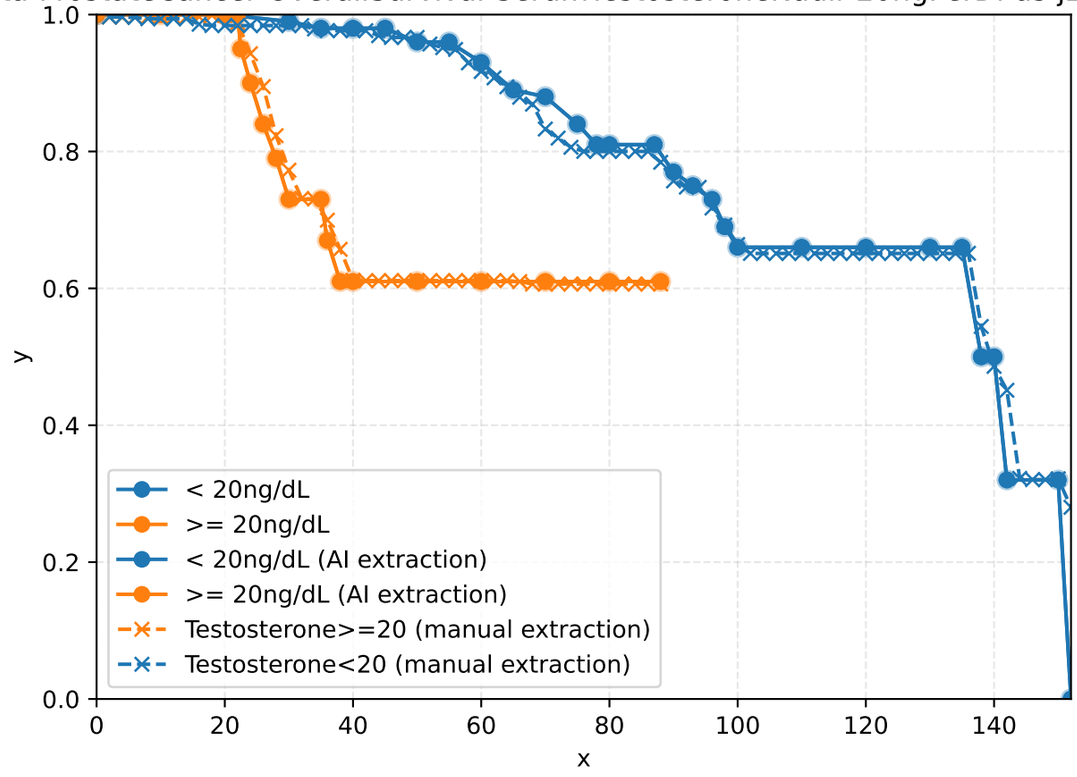
Metrics

Metric	Value
duration (s)	29
overall_nmae	0.03684
overall_integral_l1	3.32
overall_integral_l1_normalized	0.03214
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



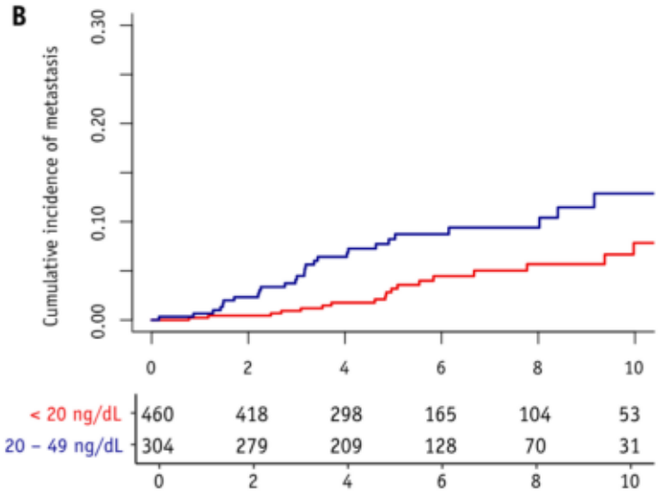
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-20ngPerDl-as-j2Uo-EdA3



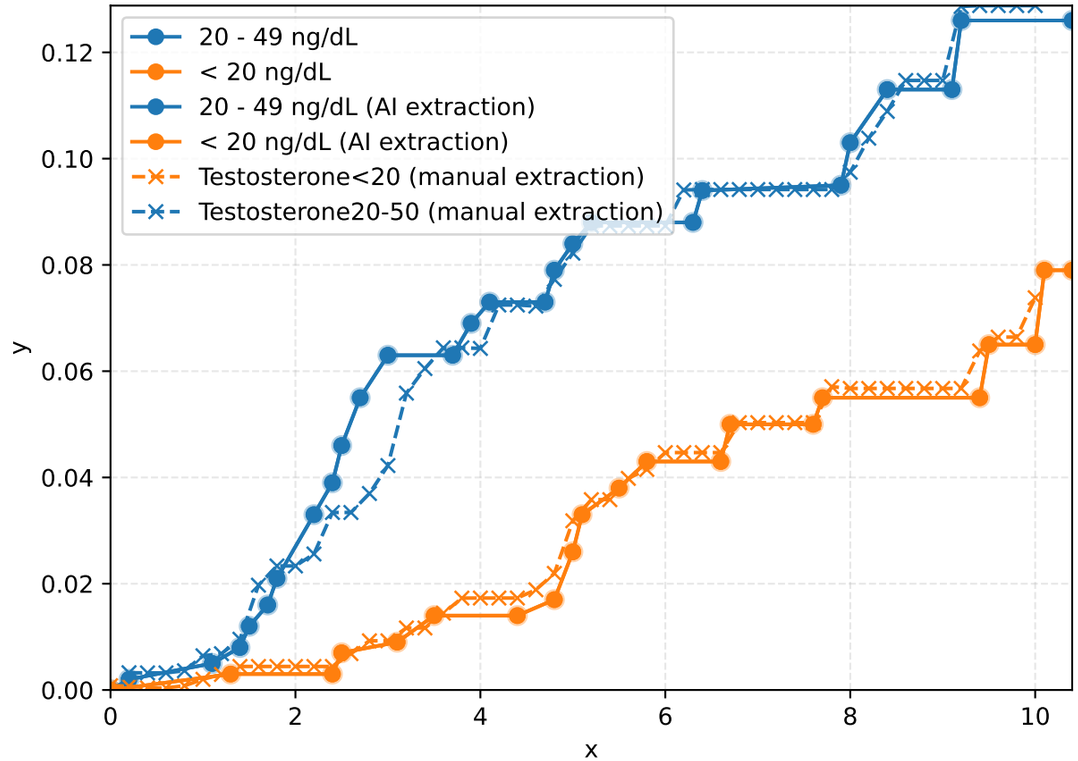
Metrics

Metric	Value
duration (s)	17
overall_nmae	0.03733
overall_integral_l1	1.581
overall_integral_l1_normalized	0.02255
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



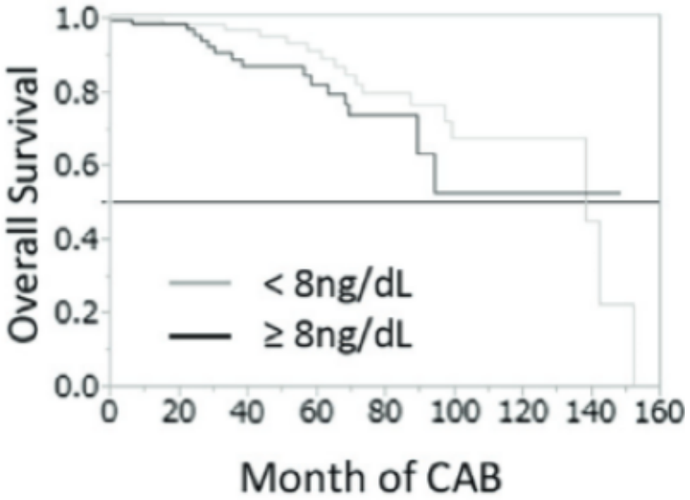
Data-ProstateCancer-Metastases-TestosteroneNadir-as-DLEW-Q14P



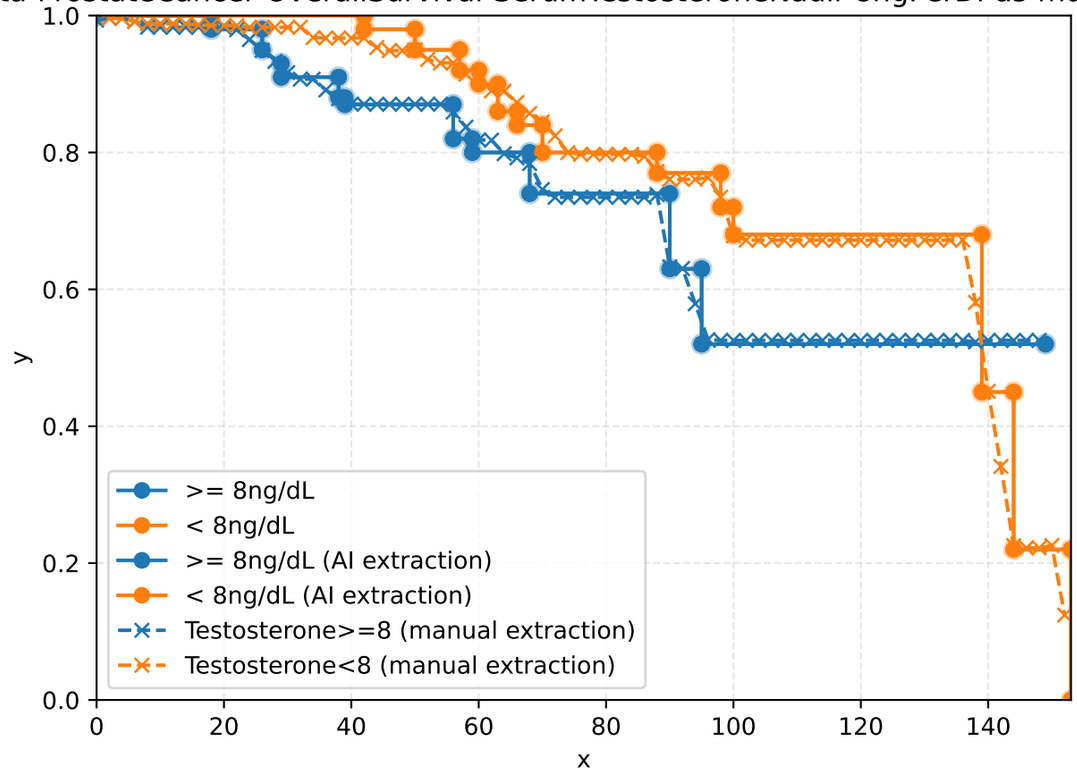
Metrics

Metric	Value
duration (s)	18
overall_nmae	0.03861
overall_integral_l1	0.02969
overall_integral_l1_normalized	0.0296
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



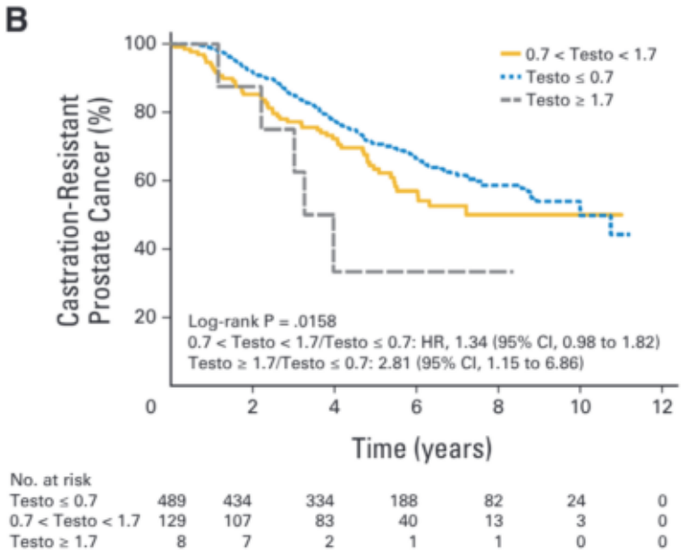
Data-ProstateCancer-OverallSurvival-SerumTestosteroneNadir-8ngPerDl-as-mu6F-pD1H



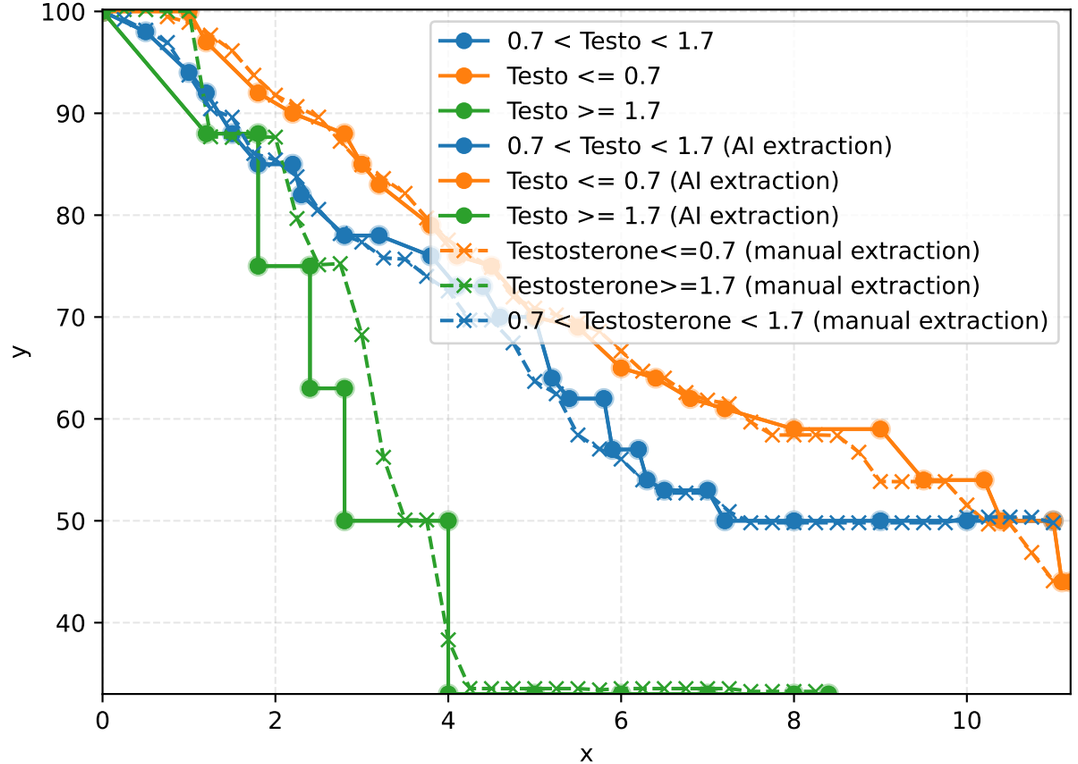
Metrics

Metric	Value
duration (s)	14
overall_nmae	0.04196
overall_integral_l1	3.841
overall_integral_l1_normalized	0.03734
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



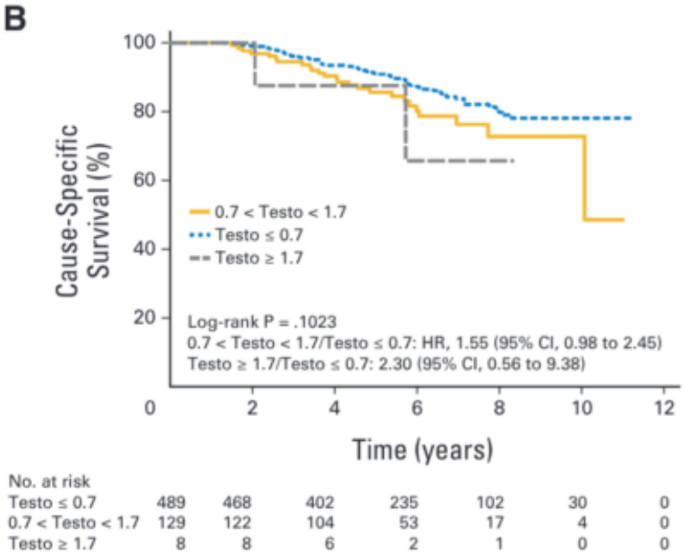
Data-ProstateCancer-TimeToCastrationResistance-TestosteroneNadir-as-KPBC-NCfw



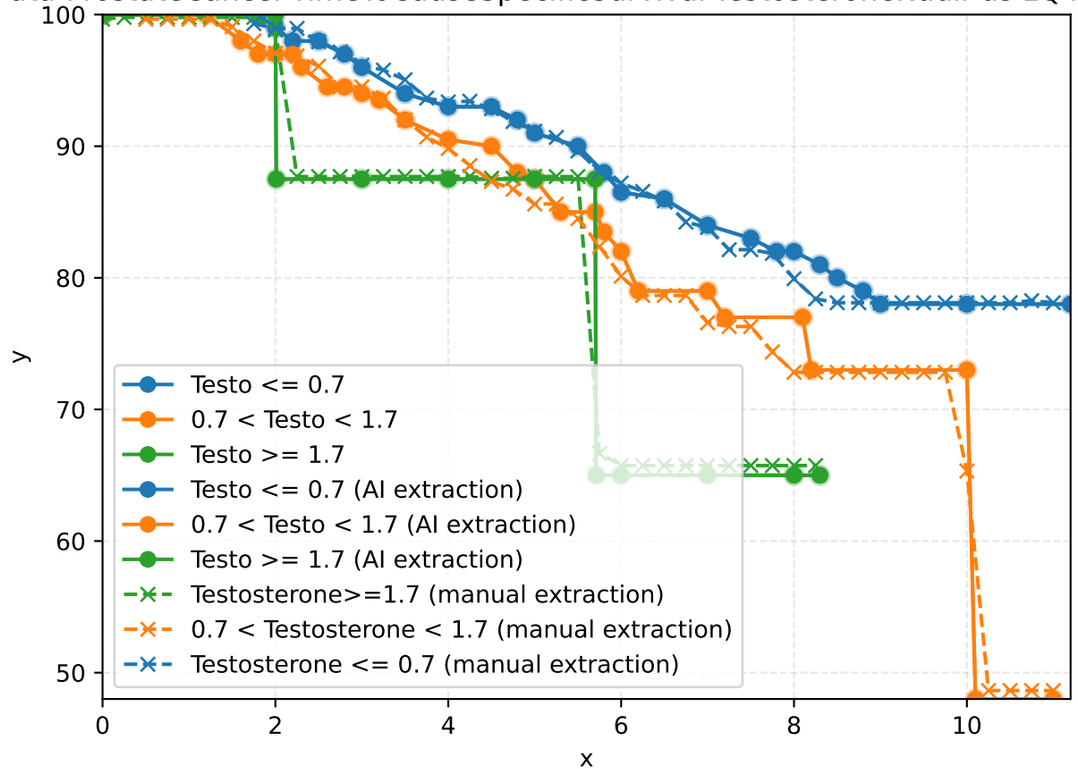
Metrics

Metric	Value
duration (s)	23
overall_nmae	0.04516
overall_integral_l1	19.81
overall_integral_l1_normalized	0.03594
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



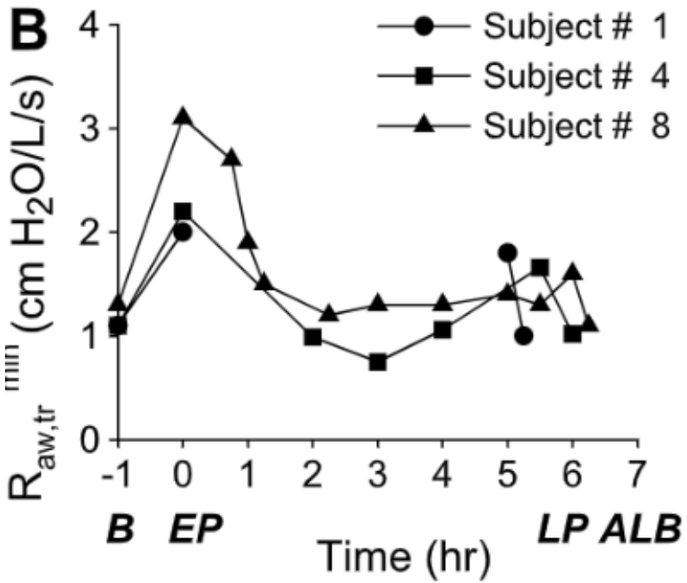
Data-ProstateCancer-TimeToCauseSpecificSurvival-TestosteroneNadir-as-zQ4P-rWVB



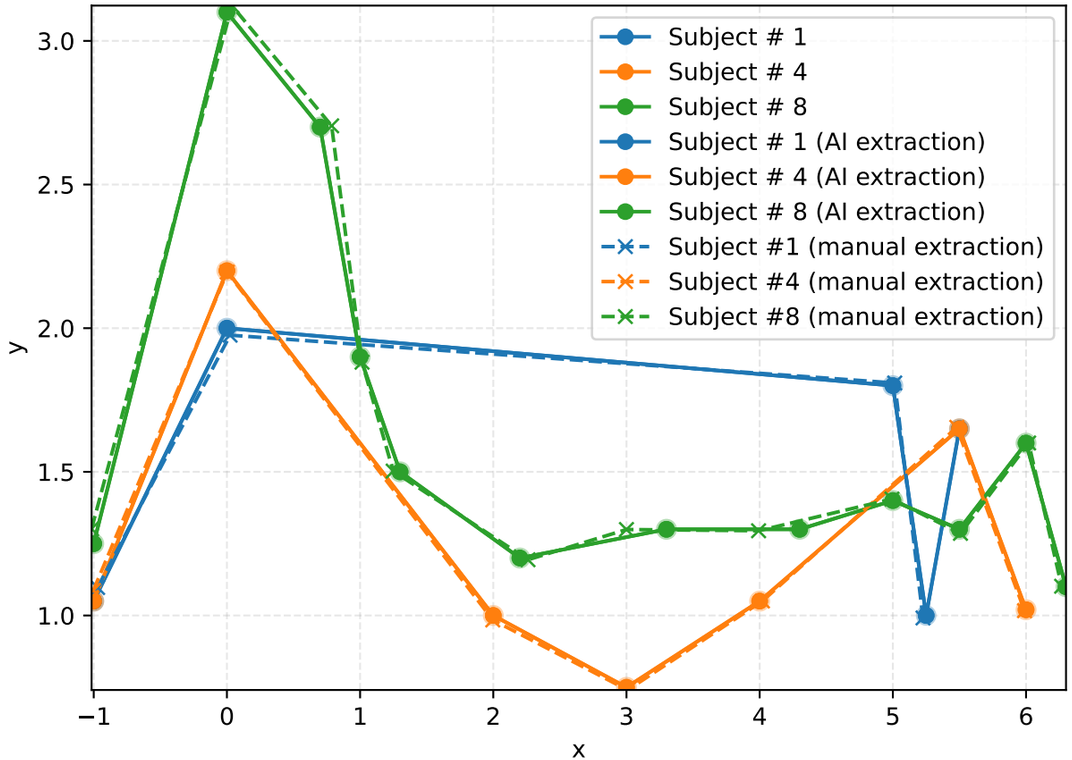
Metrics

Metric	Value
duration (s)	20
overall_nmae	0.05152
overall_integral_l1	13.82
overall_integral_l1_normalized	0.03938
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



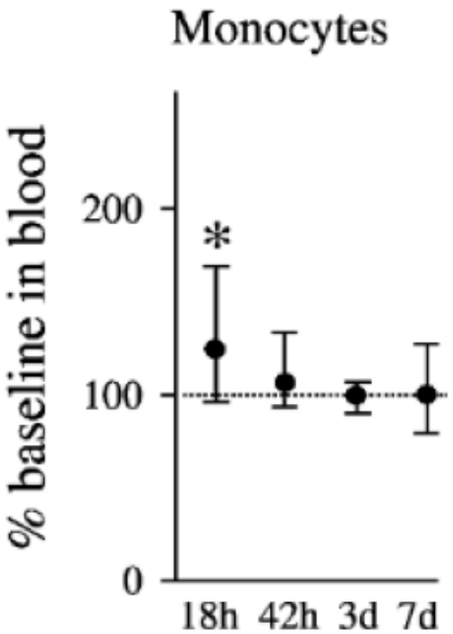
Data-as-ylJc-Vi0h



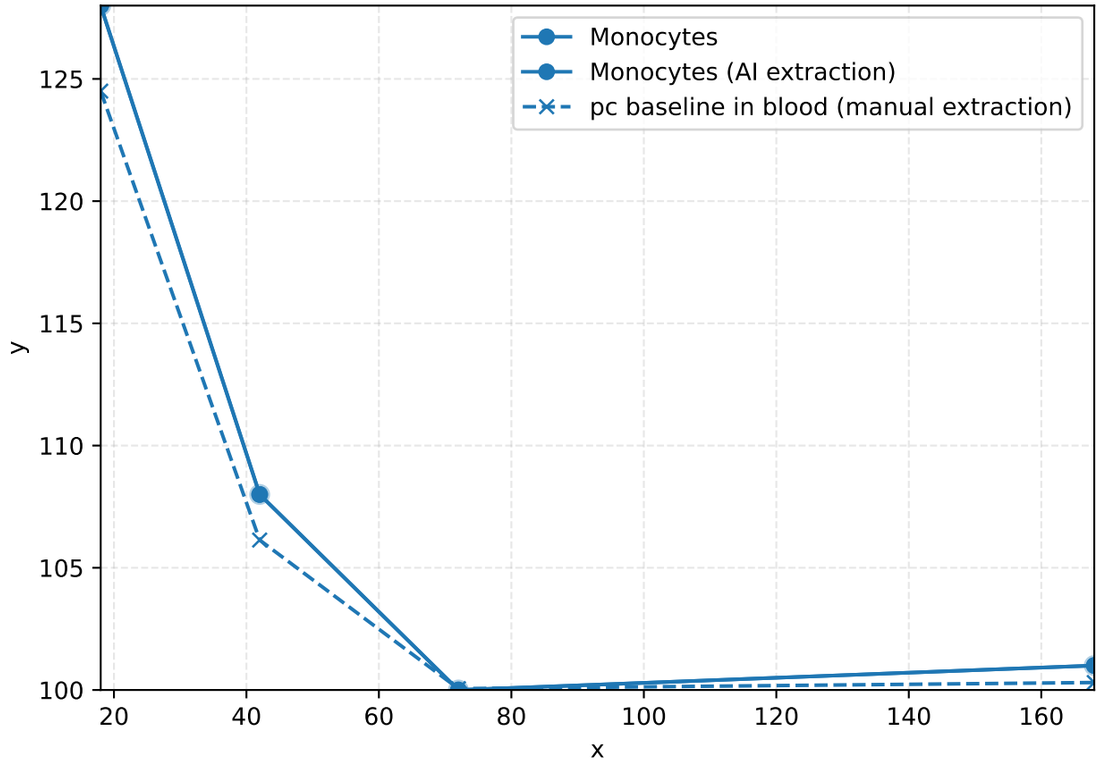
Metrics

Metric	Value
duration (s)	11
overall_nmae	0.05419
overall_integral_l1	0.1312
overall_integral_l1_normalized	0.01419
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



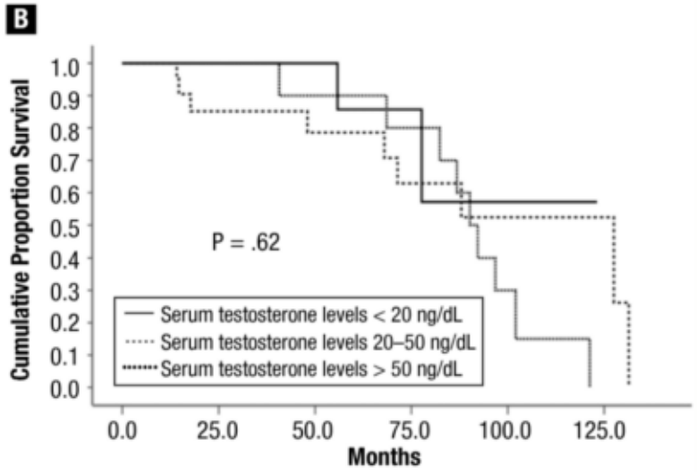
Data-as-Atl6-ihFg



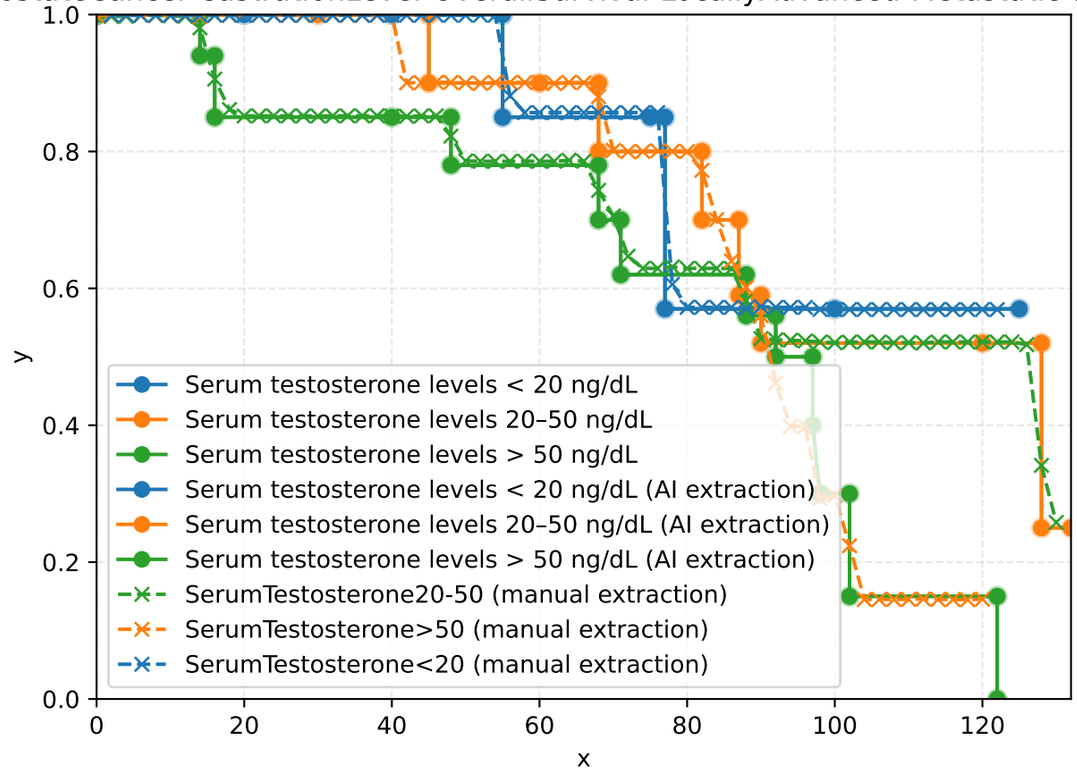
Metrics

Metric	Value
duration (s)	6
overall_nmae	0.05459
overall_integral_l1	129.2
overall_integral_l1_normalized	0.03523
overall_x_misaligned	False
extracted_series_count	1
GT_series_count	1

Source image



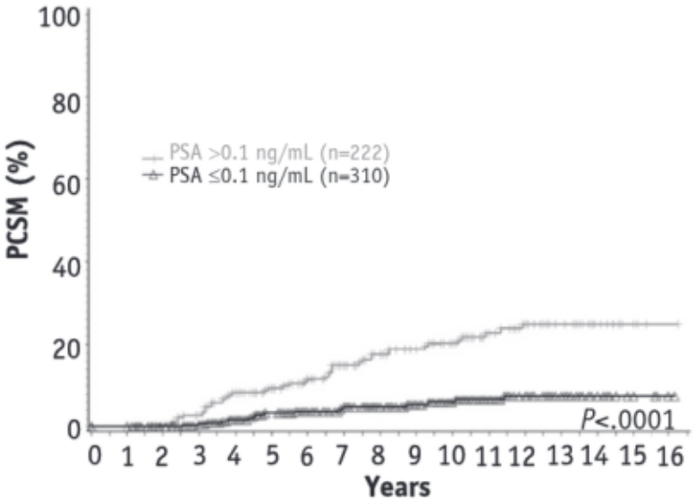
Data-ProstateCancer-CastrationLevel-OverallSurvival-LocallyAdvanced-Metastatic-as-HN9j-M4NK



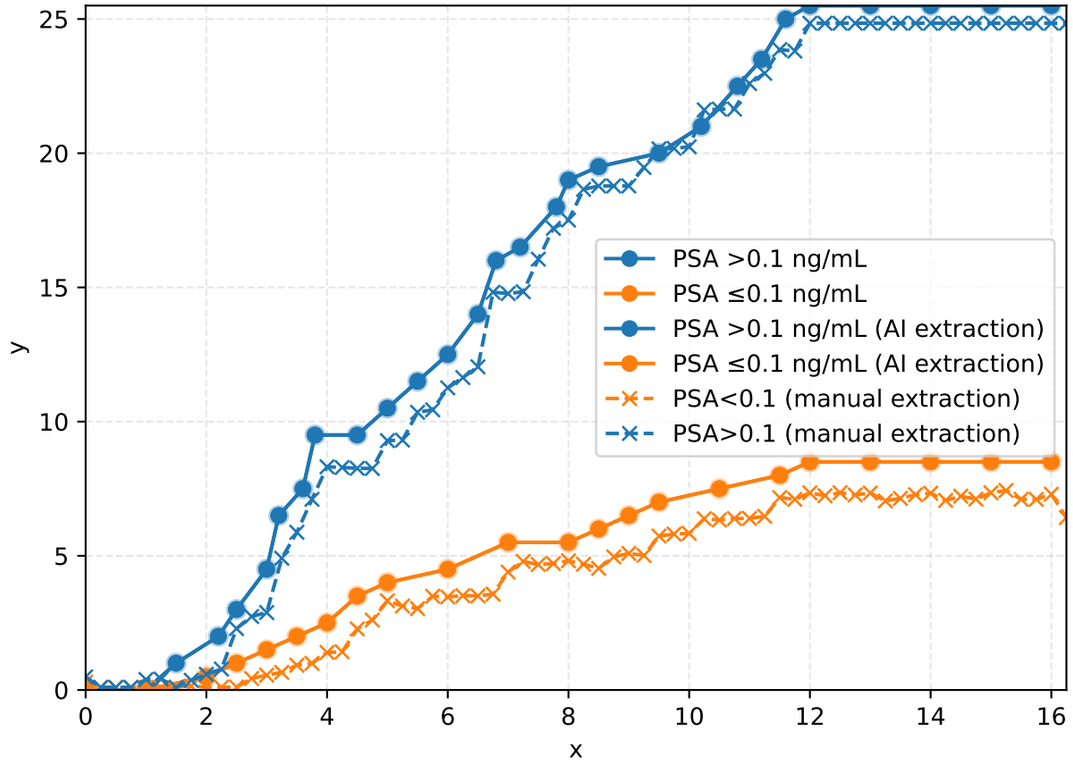
Metrics

Metric	Value
duration (s)	17
overall_nmae	0.07592
overall_integral_l1	6.529
overall_integral_l1_normalized	0.07296
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



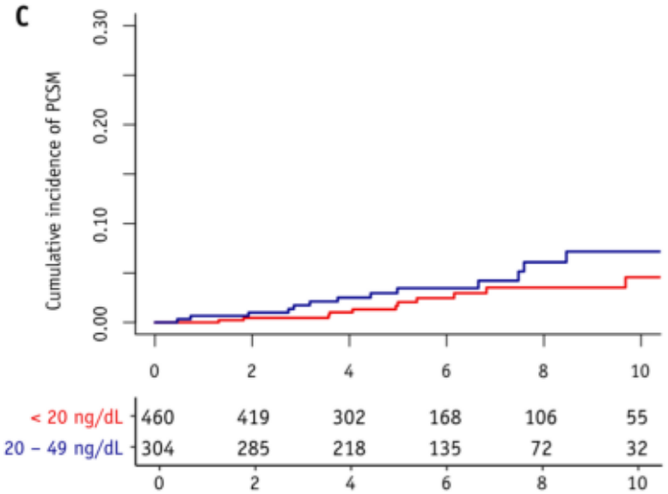
Data-Naik2016-6MonthPSA-Strat-ProstateCancerSpecificMortality-as-AXhW-IH9H



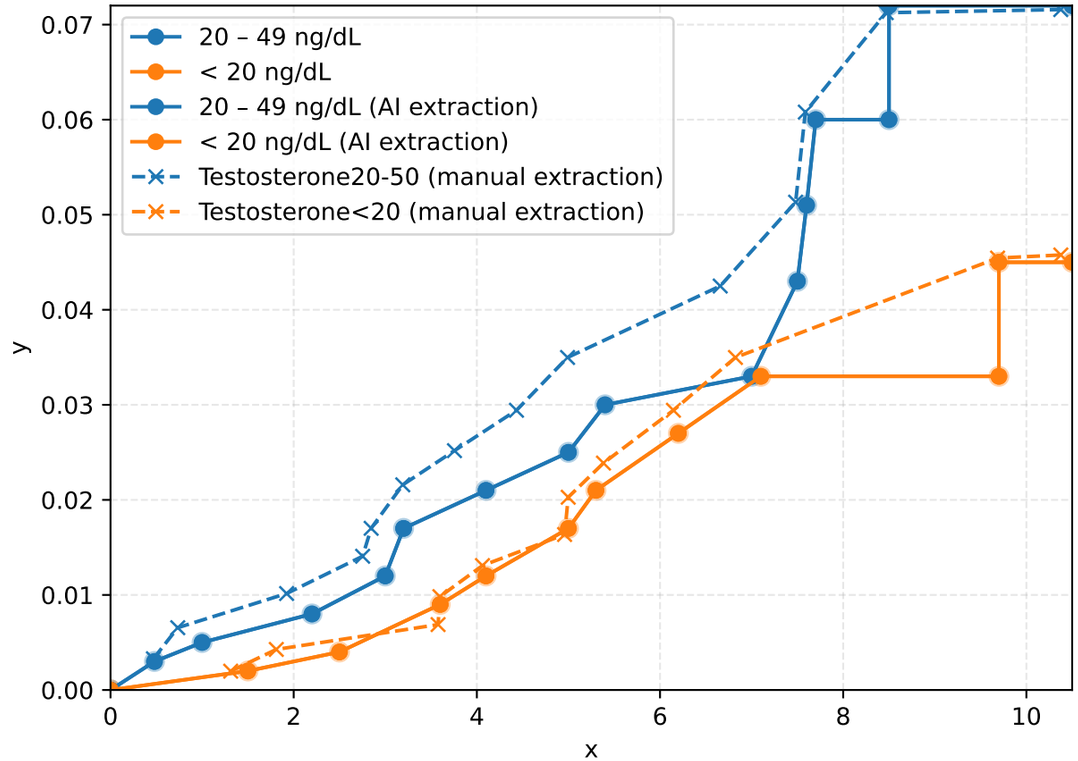
Metrics

Metric	Value
duration (s)	17
overall_nmae	0.07746
overall_integral_l1	15.01
overall_integral_l1_normalized	0.08402
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



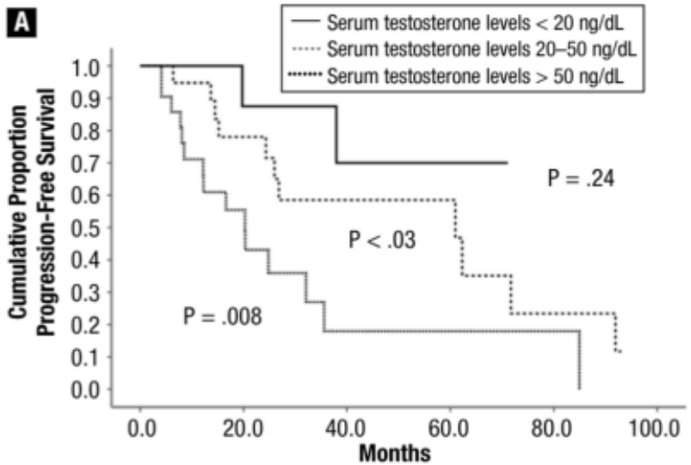
Data-ProstateCancer-ProstateCancerSpecificMortality-TestosteroneNadir-as-JAS0-KOng



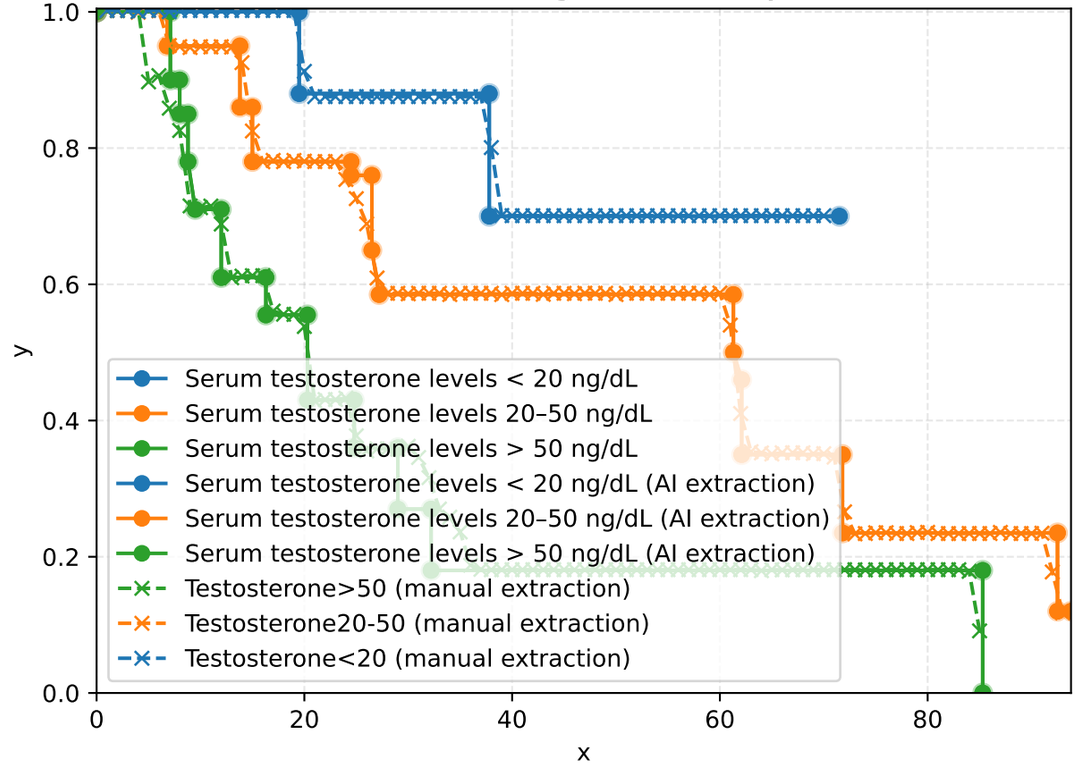
Metrics

Metric	Value
duration (s)	14
overall_nmae	0.07845
overall_integral_l1	0.04295
overall_integral_l1_normalized	0.07975
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



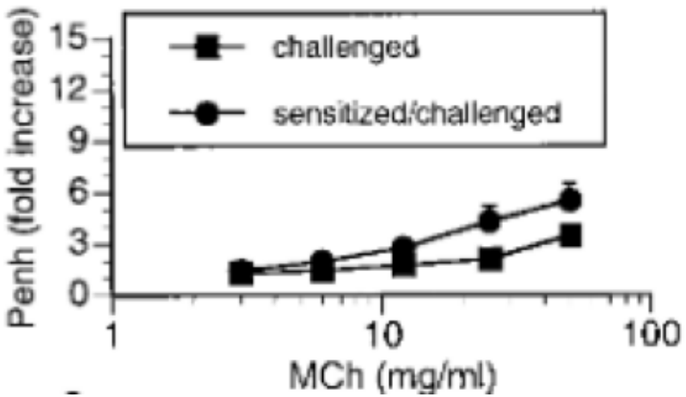
Data-ProstateCancer-CastrationLevel-TimeToProgression-LocallyAdvanced-Metastatic-as-aWvR-2wlo



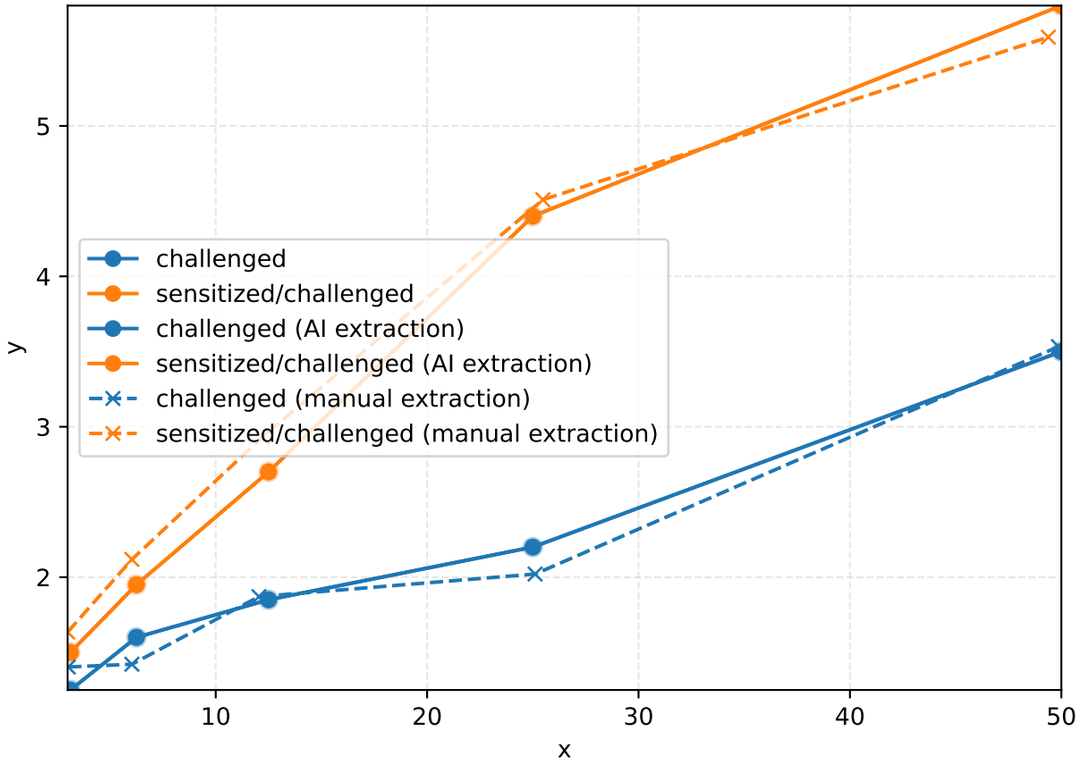
Metrics

Metric	Value
duration (s)	24
overall_nmae	0.08792
overall_integral_l1	1.949
overall_integral_l1_normalized	0.03427
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



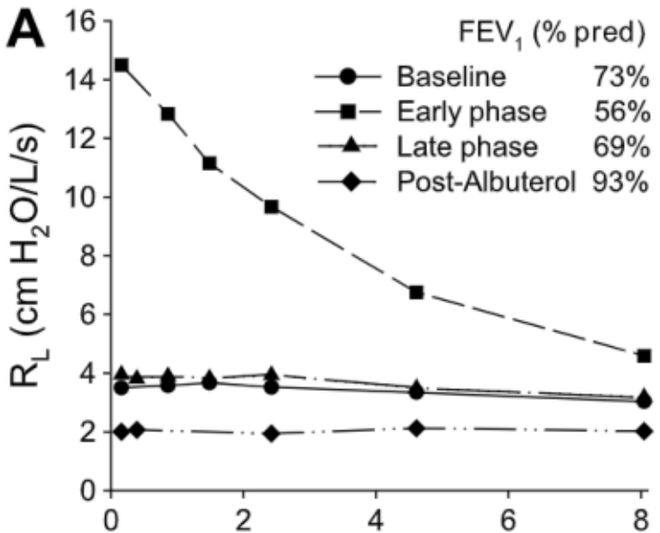
Data-as-CdWI-s6OK



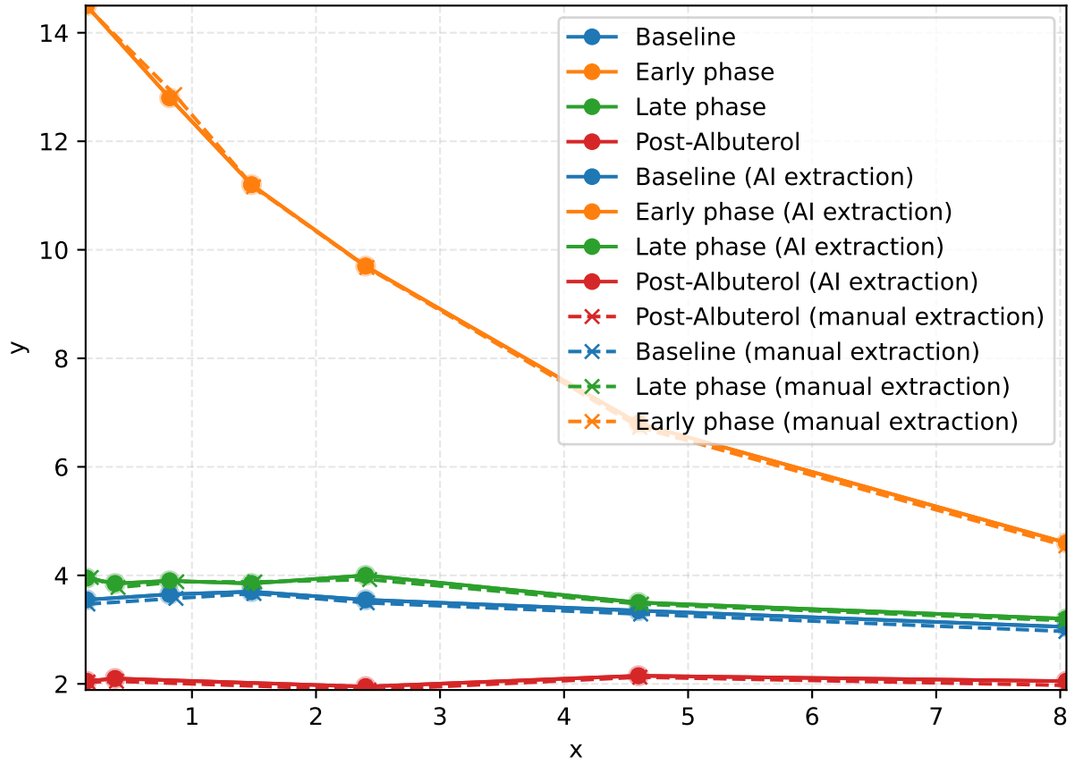
Metrics

Metric	Value
duration (s)	7
overall_nmae	0.0972
overall_integral_l1	3.181
overall_integral_l1_normalized	0.02283
overall_x_misaligned	False
extracted_series_count	2
GT_series_count	2

Source image



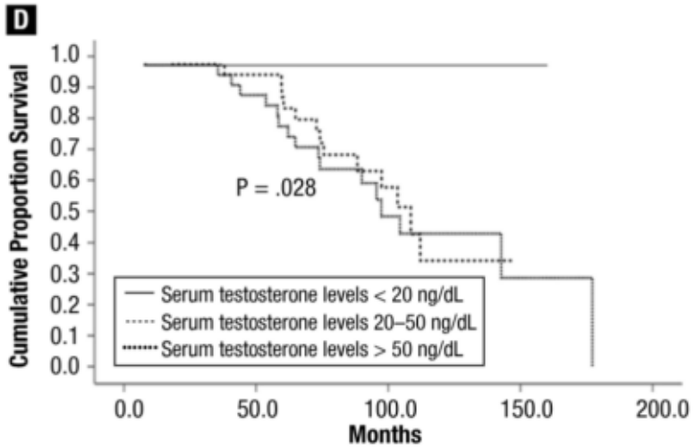
Data-as-y2nv-TPY1



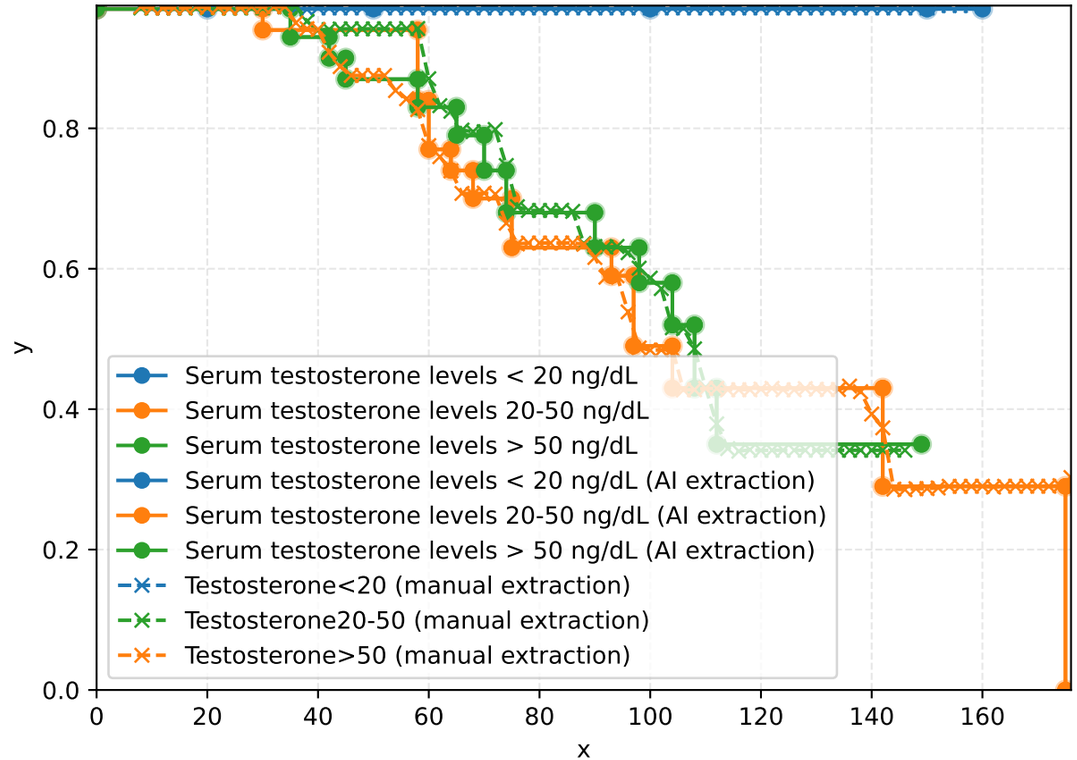
Metrics

Metric	Value
duration (s)	13
overall_nmae	0.1028
overall_integral_l1	0.2038
overall_integral_l1_normalized	0.04647
overall_x_misaligned	False
extracted_series_count	4
GT_series_count	4

Source image



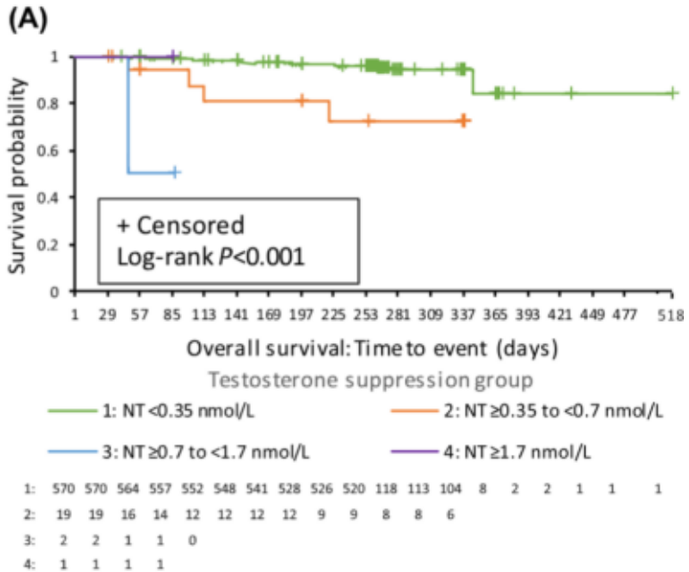
Data-ProstateCancer-CastrationLevel-OverallSurvival-BiochemicalRecurrence-as-KBNL-4YCX



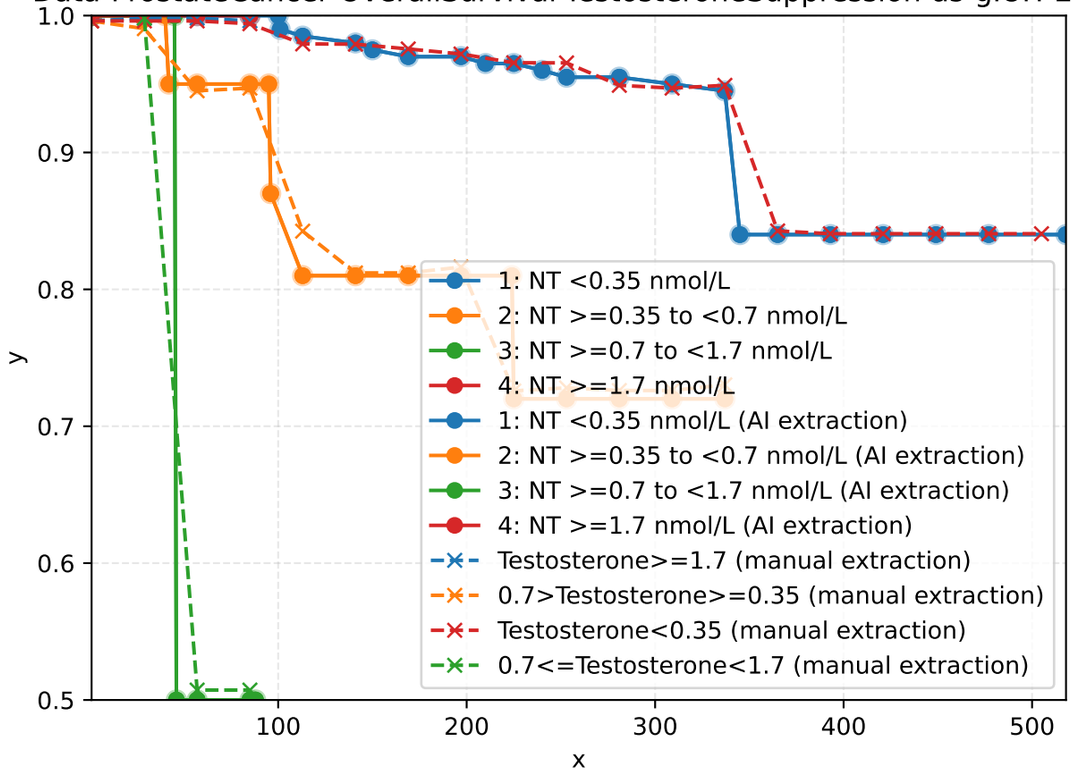
Metrics

Metric	Value
duration (s)	18
overall_nmae	0.2131
overall_integral_l1	2.556
overall_integral_l1_normalized	0.1309
overall_x_misaligned	False
extracted_series_count	3
GT_series_count	3

Source image



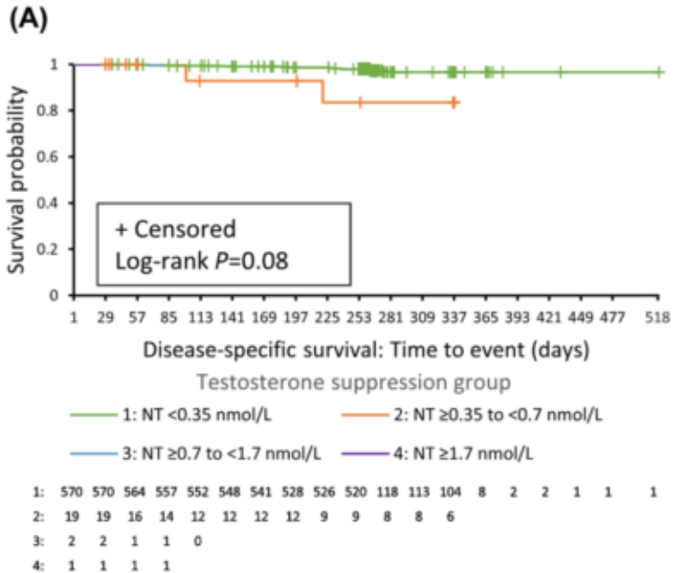
Data-ProstateCancer-OverallSurvival-TestosteroneSuppression-as-gIUH-EPAn



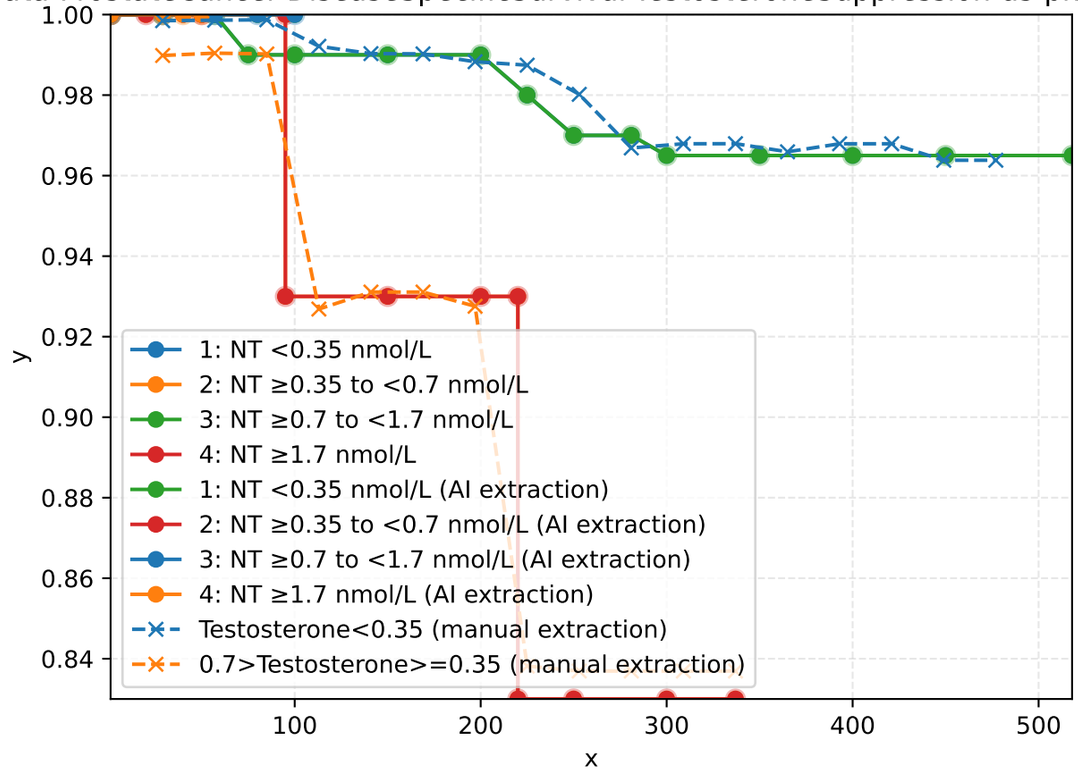
Metrics

Metric	Value
duration (s)	18
overall_nmae	0.4457
overall_integral_l1	1.598
overall_integral_l1_normalized	0.3413
overall_x_misaligned	True
extracted_series_count	4
GT_series_count	4

Source image



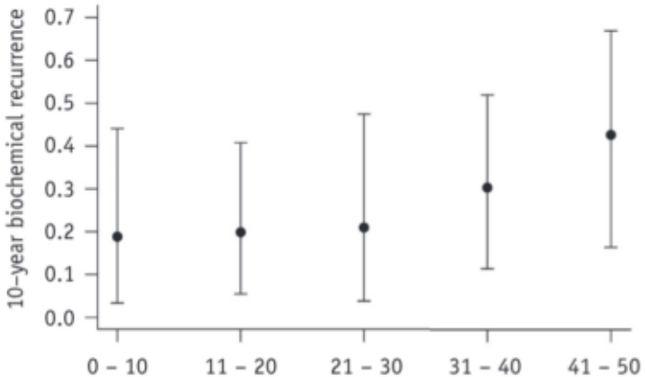
Data-ProstateCancer-DiseaseSpecificSurvival-TestosteroneSuppression-as-pk2y-g1My



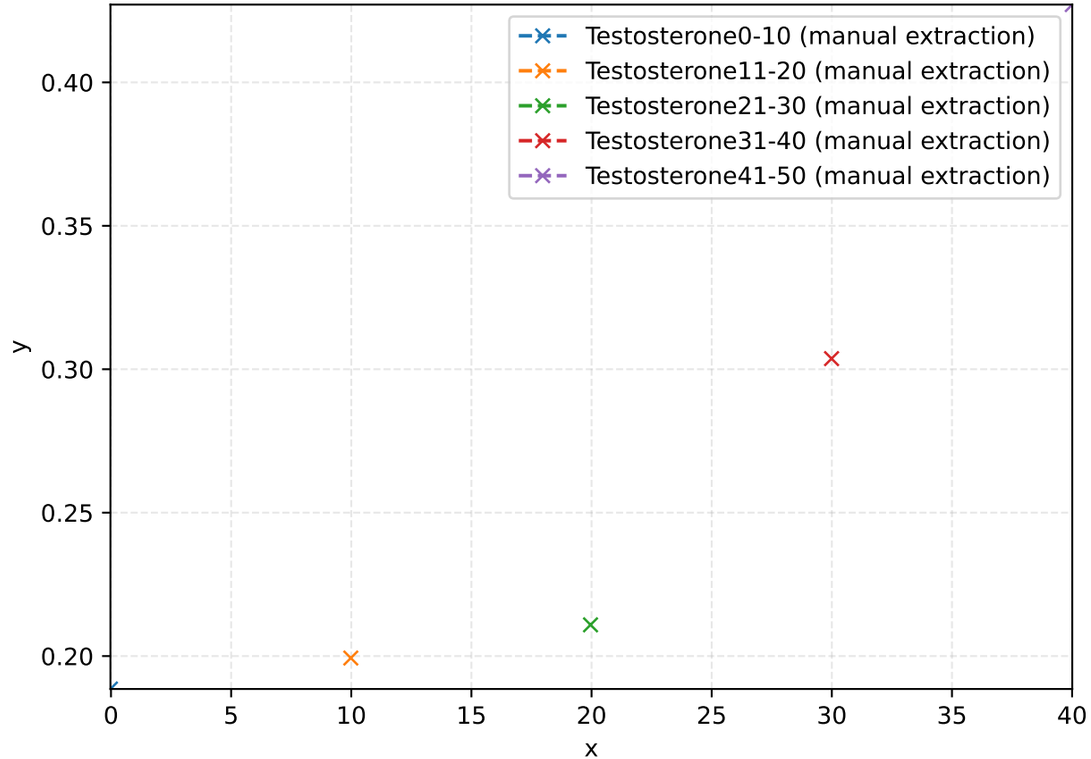
Metrics

Metric	Value
duration (s)	13
overall_nmae	0.7234
overall_integral_l1	0.1708
overall_integral_l1_normalized	0.05922
overall_x_misaligned	True
extracted_series_count	4
GT_series_count	2

Source image



Data-ProstateCancer-10YearBiochemicalRecurrence-TestosteroneNadir-as-FRp3-CyQW



Metrics

Metric	Value
duration (s)	—
overall_nmae	—
overall_integral_l1	—
overall_integral_l1_normalized	—
overall_x_misaligned	—
extracted_series_count	—
GT_series_count	—